

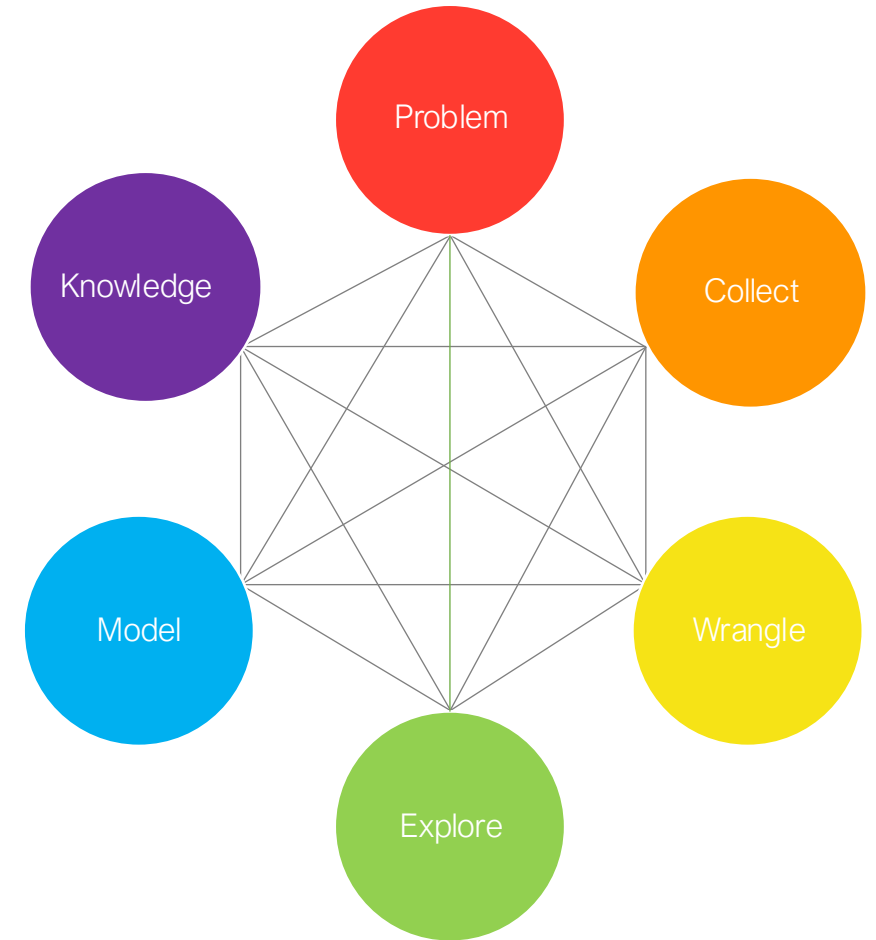
GEOG0114: PRINCIPLES OF SPATIAL ANALYSIS

WEEK 1: INTRODUCTION TO SPATIAL ANALYSIS & DATA SCIENCE

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Lecturer in Social and Geographic Data Science
UCL Geography

Contents

- What will we learn in GEOG0114?
- What is Spatial Statistics?
- What are Spatial Data & Features?
- What is Geographic Information Systems?



What will we learn?

What will we learn?

- To introduce you to key principles of spatial statistics
- Provide an introduction and knowledge of methods for exploring various types of spatial data (i.e., point, areal, line segments and gridded/pixeled data)
- You will know how to adopt various spatial analytical techniques for testing out hypothesis, and for addressing problems related to social phenomena and its spatial components.
- You will learn to how to apply various families of spatial models (e.g., geographic weighted regressions, Global & Local Moran's I, Kriging and many more) for making spatial predictions and studying patterns of associations between risk factors and outcomes
- You will gain programming skills for carry out data managing, geoprocessing and analysis of spatial data using the software package R/RStudio. You will gain expert knowledge on how to use R/RStudio as a GIS software to perform high-level map visualization

Structure of course

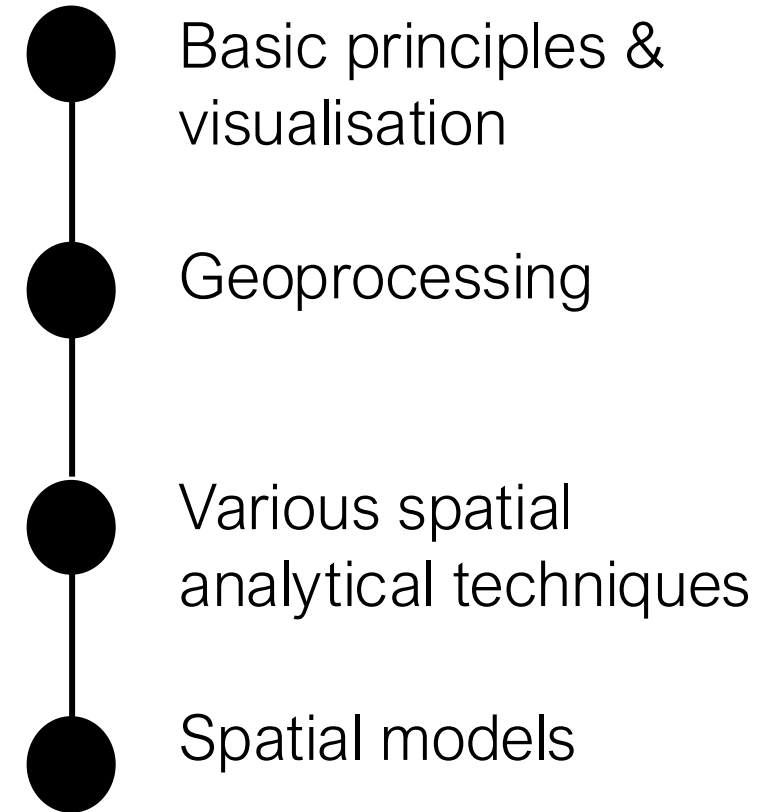
The first part of the module provides foundational knowledge for performing geospatial data analysis with various types of spatial data. In the second part of the module, we will focus on key concepts and methods for dealing gridded data using Raster-based analysis. The third part of the module, we will introduce you to more specialized techniques with real-world applications; and lastly, we will various spatial modelling techniques for evidence-based research.

Part One (Week 1-3): Foundation and theory (i.e., representation of spatial data, geoprocessing and spatial autocorrelation)

Part Two (Week 4-6): Raster-based analysis (i.e., Suitability mapping e.g., AHPs and Niches; and geostatistical modeling using Kriging)

Part Three (Week 7, 8 and 9): Spatial Models e.g., Lag and Error Models, GWRs, Spatial Risk Models.

Part Four (Week 10): Revision



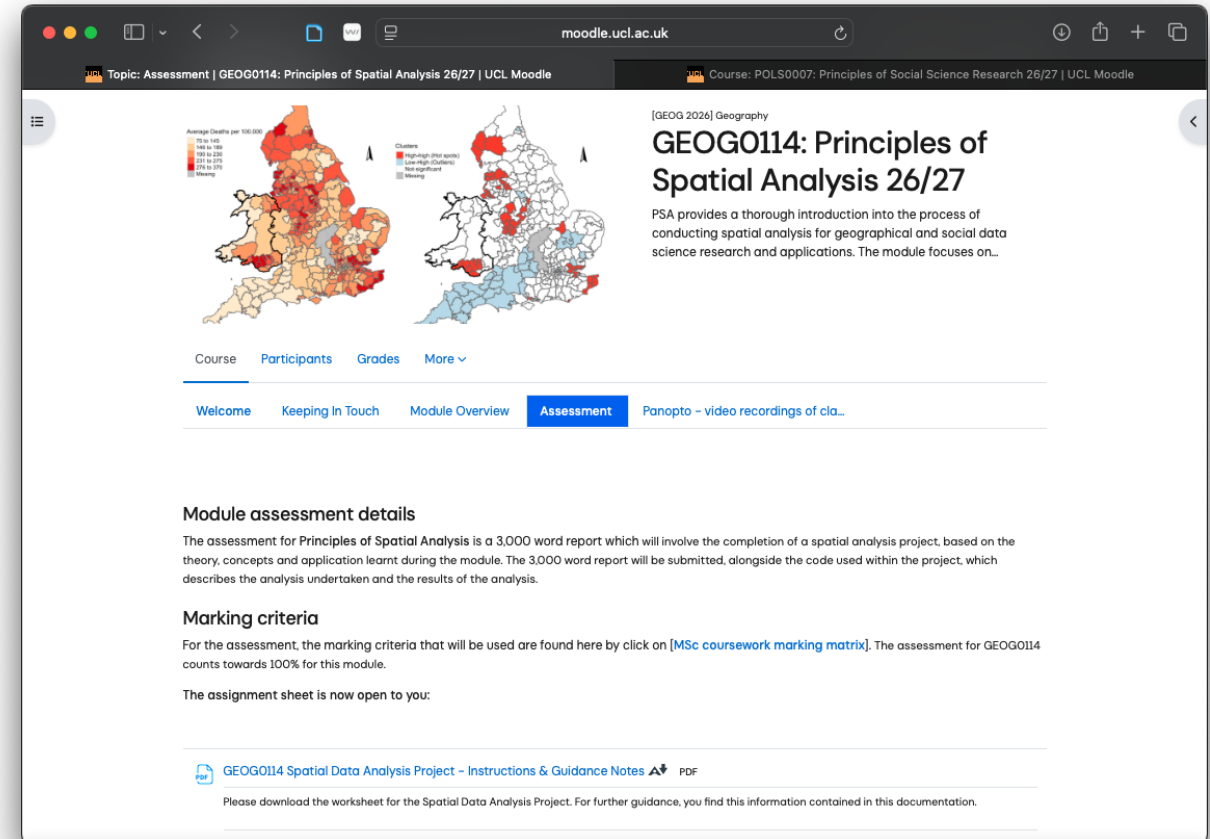
Breakdown of what will we be learning

Week	Weekly Topics
01	Spatial Analysis for Data Science (AM)
02	Graphical Representation of Spatial Data (AM)
03	Spatial Autocorrelation (AM)
04	Suitability Mapping: Part I (Analytical Hierarchy Process) (AM)
05	Suitability Mapping: Part II (Ecological Niche Models) (AM)
Reading Week	
06	Geostatistics using Kriging (AM)
07	Geodemographic Classification (JVD)
08	Accessibility Analysis (JVD)
09	Spatial Models: Part I (Spatial Lag & Error Models) (AM)
10	Spatial Models: Part II (Geographically Weighted Regression) (AM)

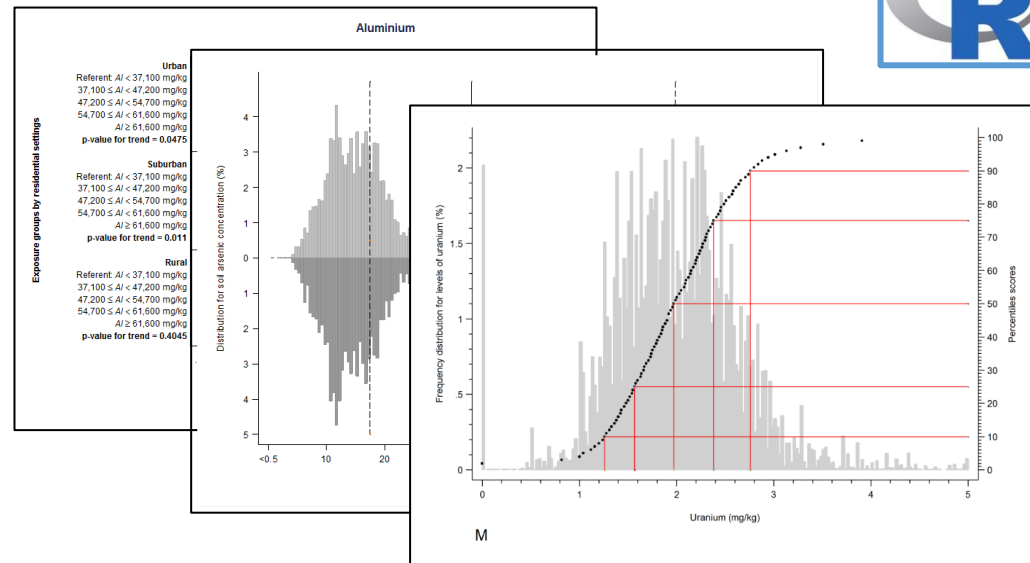
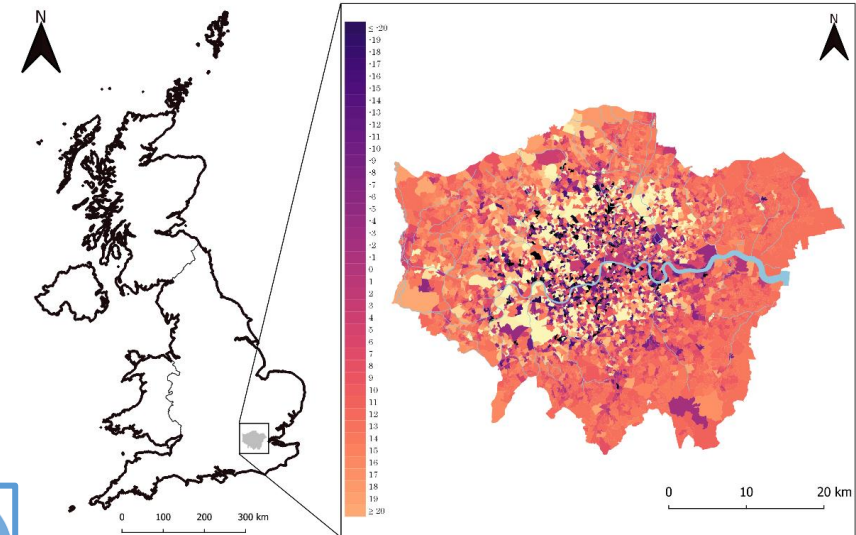
Computer practical sessions - Please bring your own laptops with you to the computer practicals on Thursday @03:00pm-05:00pm

Coursework & Assessment

- What is the assessment?
 - Spatial data analysis project which is worth 100%
 - It is a report (max 3,000 words) + code (R).
- You can check the coursework out right now under the Assessment section, and start thinking about the topic for your project.
- The deadline for the coursework is in the 1st week of **January (Term 2)**. The submission instructions will be available on Moodle.



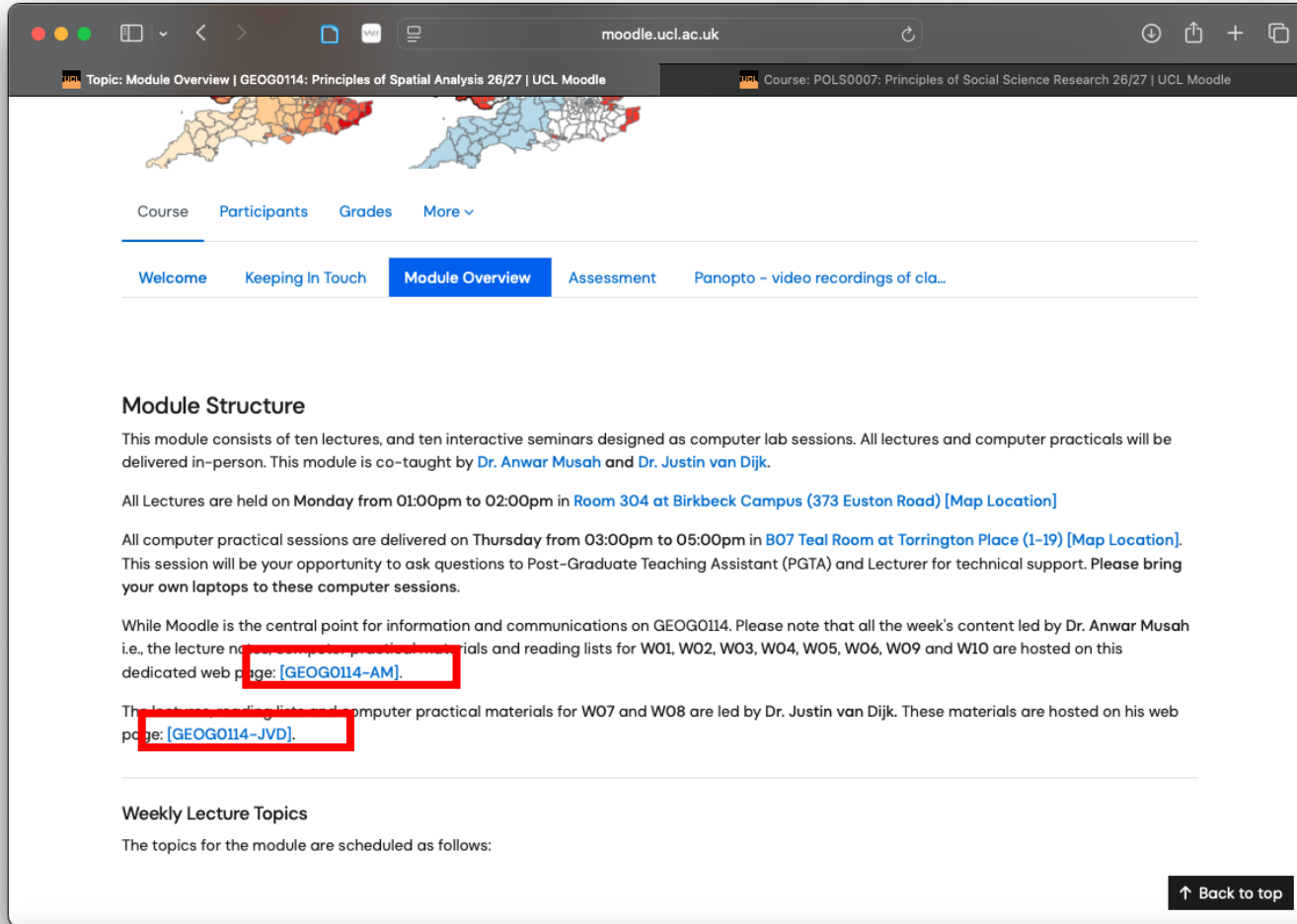
A word cloud featuring various statistics-related terms. The most prominent words are 'data', 'statistical', and 'statisti'. Other visible words include 'analysis', 'models', 'inference', 'quantitative', 'deviation', 'research', 'coefficient', 'regression', 'learning', 'computing', 'generalized', 'linear', 'bayesian', 'probability', 'modeling', 'maximum', 'sampling', 'estimation', 'random', 'machine', 'workshops', 'simulation', 'causal', 'equation', 'consulting', 'series', 'covariate', 'duration', 'variance', 'distribution', 'spatial', 'visualization', 'methods', 'predictive', 'normal', 'time', 'function', 'parameter', 'management', 'trend', 'likelihood', 'station', 'analytics', and 'research'. The words are in various colors (blue, orange, green, red) and orientations (horizontal, vertical).



```

181 raster_file <- raster(file)
182 recife_temperature_cropped <- crop(raster_file, recife_extent)
183 recife_temperature_masked <- mask(recife_temperature_cropped, bra_recife_outline)
184 recife_temperature_masked <- projectRaster(recife_temperature_masked, crs=pcrs)
185 recife_temp_aggr <- extract(recife_temperature_masked, bra_recife_areas, fun=mean, d
186 recife_temp_aggr$districtID<-bra_recife_areas$ID
187 colnames(recife_temp_aggr)[1] <- "fid"
188 colnames(recife_temp_aggr)[2] <- "temperature"
189 colnames(recife_temp_aggr)[3] <- "district_id"
190 recife_temp_aggr$year <- i
191 recife_temp_aggr$month <- j
192 recife_temperature <- recife_temp_aggr[,c(1,3,4,5,2)]
193 }
194 else {
195   file <- paste0("/Users/anwarmusah/Desktop/AM_Zika2019/Data/Brazil/Climatic/Temperature
196 raster_file <- raster(file)
197 recife_temperature_cropped <- crop(raster_file, recife_extent)
198 recife_temperature_masked <- mask(recife_temperature_cropped, bra_recife_outline)
199 recife_temperature_masked <- projectRaster(recife_temperature_masked, crs=pcrs)

```



moodle.ucl.ac.uk

Topic: Module Overview | GEOG0114: Principles of Spatial Analysis 26/27 | UCL Moodle

Course: POLS0007: Principles of Social Science Research 26/27 | UCL Moodle

Course Participants Grades More

Welcome Keeping In Touch **Module Overview** Assessment Panopto – video recordings of cla...

Module Structure

This module consists of ten lectures, and ten interactive seminars designed as computer lab sessions. All lectures and computer practicals will be delivered in-person. This module is co-taught by [Dr. Anwar Musah](#) and [Dr. Justin van Dijk](#).

All Lectures are held on Monday from 01:00pm to 02:00pm in [Room 304 at Birkbeck Campus \(373 Euston Road\)](#) [\[Map Location\]](#)

All computer practical sessions are delivered on Thursday from 03:00pm to 05:00pm in [B07 Teal Room at Torrington Place \(1-19\)](#) [\[Map Location\]](#). This session will be your opportunity to ask questions to Post-Graduate Teaching Assistant (PGTA) and Lecturer for technical support. Please bring your own laptops to these computer sessions.

While Moodle is the central point for information and communications on GEOG0114. Please note that all the week's content led by Dr. Anwar Musah i.e., the lecture notes, computer practical materials and reading lists for W01, W02, W03, W04, W05, W06, W09 and W10 are hosted on this dedicated web page: [\[GEOG0114-AM\]](#).

The lecture notes, reading lists and computer practical materials for W07 and W08 are led by Dr. Justin van Dijk. These materials are hosted on his web page: [\[GEOG0114-JVD\]](#).

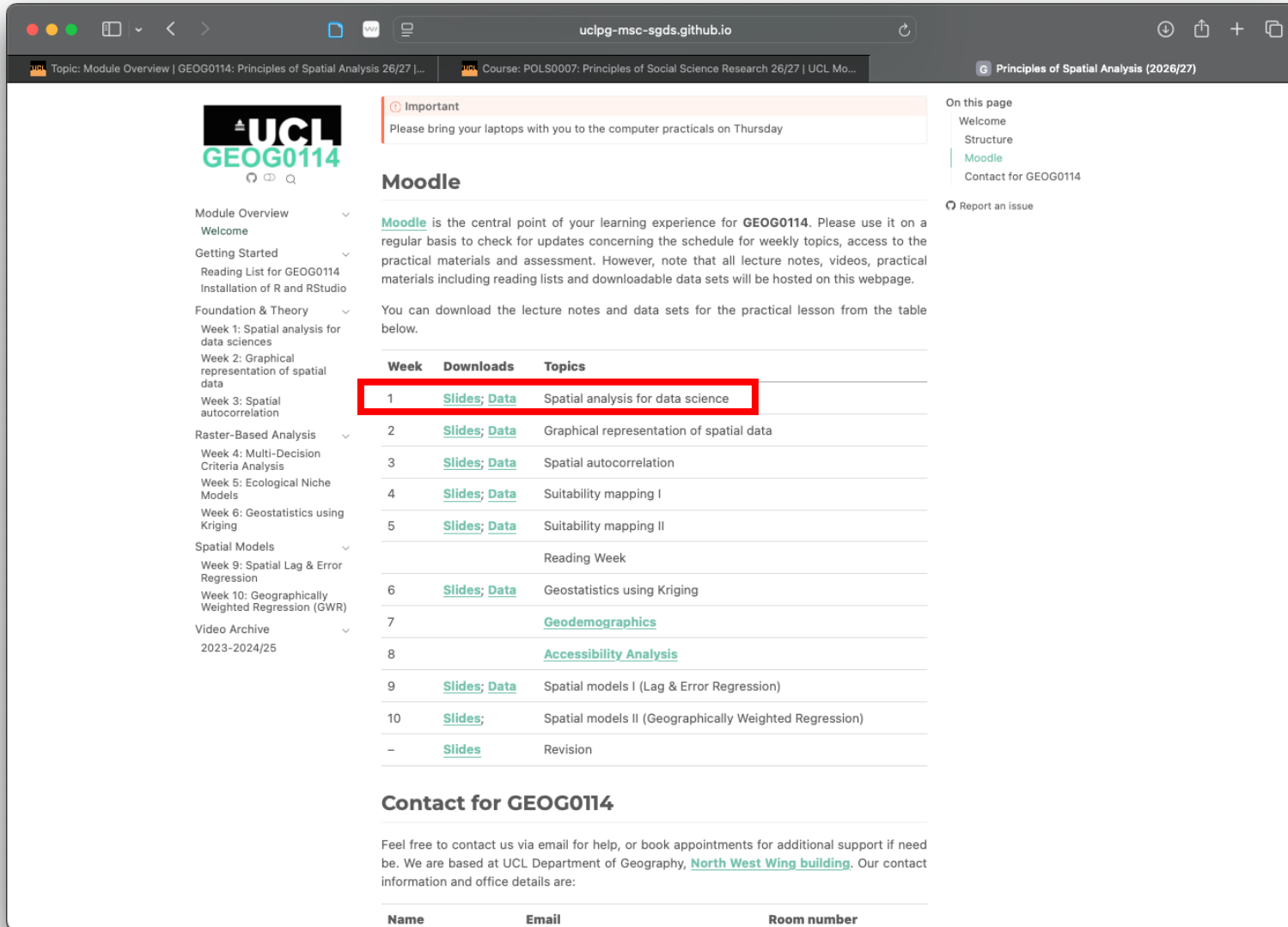
Weekly Lecture Topics

The topics for the module are scheduled as follows:

[↑ Back to top](#)

- Moodle for GEOG0114 is central point for course and assignment information
[\[https://moodle.ucl.ac.uk/\]](https://moodle.ucl.ac.uk/)
- **Module Overview** section is where you can see the timetable and access **GEOG0114 GitHub website**
- Please click on any of these links to be teleported to the **GEOG0114 GitHub website**.
- Here is where the practical and lecture note materials are hosted. Please download as soon as you gain access.
- All lecture notes and computer practical materials are available to you. I'll try to keep the materials and codes updated as much as possible

https://uclpg-msc-sgds.github.io/GEOG0114/00-module_overview.html



Topic: Module Overview | GEOG0114: Principles of Spatial Analysis 26/27 | ... Course: POLS0007: Principles of Social Science Research 26/27 | UCL Mo... Principles of Spatial Analysis (2026/27)

UCL GEOG0114

Module Overview
Welcome
Getting Started
Reading List for GEOG0114
Installation of R and RStudio
Foundation & Theory
Week 1: Spatial analysis for data sciences
Week 2: Graphical representation of spatial data
Week 3: Spatial autocorrelation
Raster-Based Analysis
Week 4: Multi-Decision Criteria Analysis
Week 5: Ecological Niche Models
Week 6: Geostatistics using Kriging
Spatial Models
Week 9: Spatial Lag & Error Regression
Week 10: Geographically Weighted Regression (GWR)
Video Archive
2023-2024/25

Important
Please bring your laptops with you to the computer practicals on Thursday

Moodle
Moodle is the central point of your learning experience for GEOG0114. Please use it on a regular basis to check for updates concerning the schedule for weekly topics, access to the practical materials and assessment. However, note that all lecture notes, videos, practical materials including reading lists and downloadable data sets will be hosted on this webpage.
You can download the lecture notes and data sets for the practical lesson from the table below.

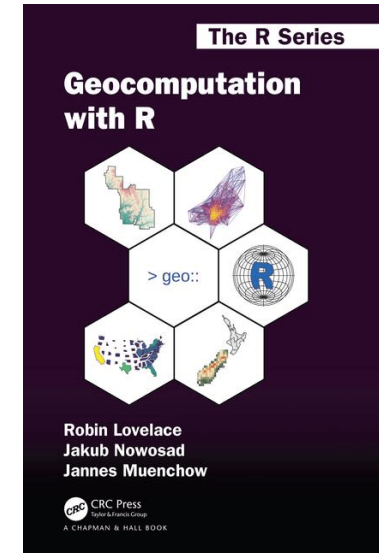
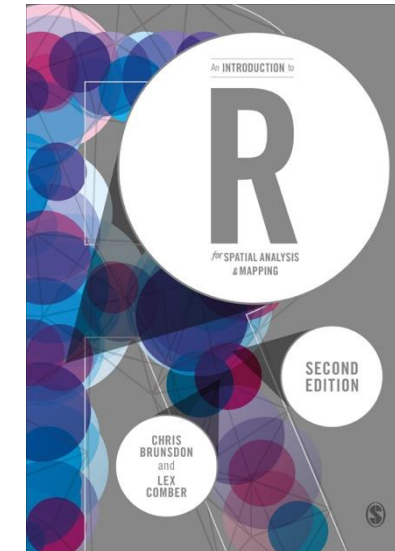
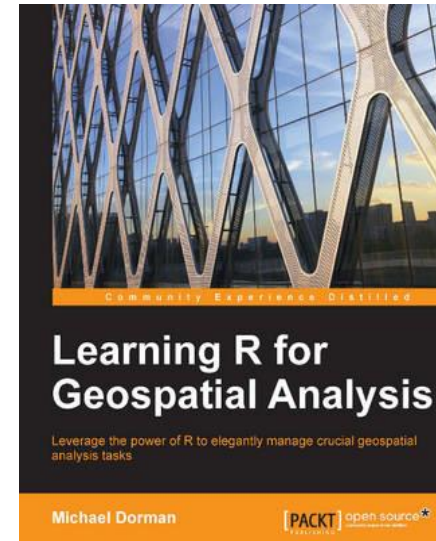
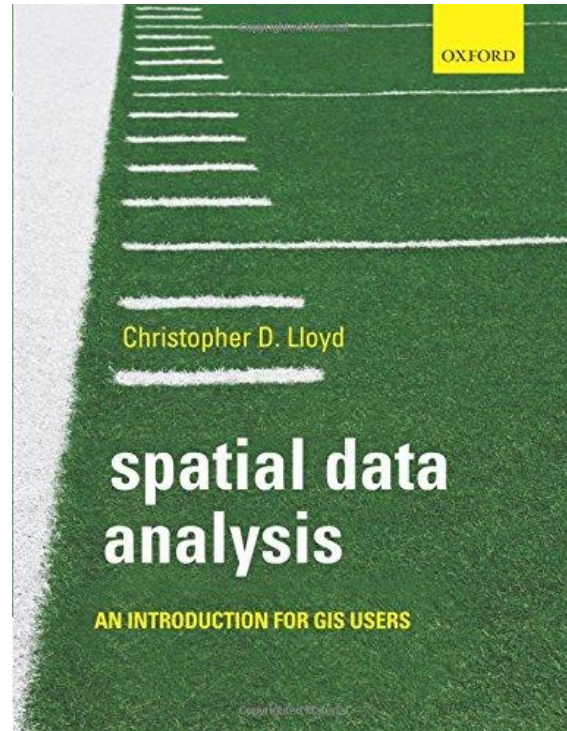
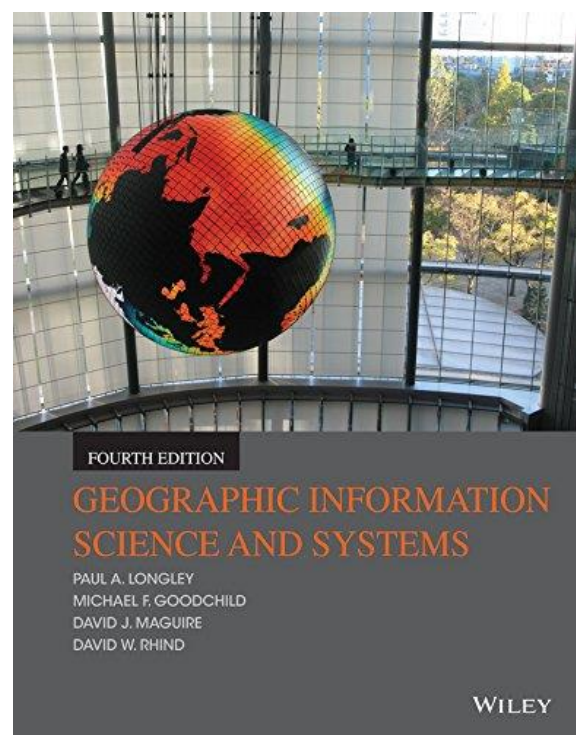
Week	Downloads	Topics
1	Slides ; Data	Spatial analysis for data science
2	Slides ; Data	Graphical representation of spatial data
3	Slides ; Data	Spatial autocorrelation
4	Slides ; Data	Suitability mapping I
5	Slides ; Data	Suitability mapping II
		Reading Week
6	Slides ; Data	Geostatistics using Kriging
7		Geodemographics
8		Accessibility Analysis
9	Slides ; Data	Spatial models I (Lag & Error Regression)
10	Slides ;	Spatial models II (Geographically Weighted Regression)
-	Slides	Revision

Contact for GEOG0114
Feel free to contact us via email for help, or book appointments for additional support if need be. We are based at UCL Department of Geography, [North West Wing building](#). Our contact information and office details are:

Name	Email	Room number
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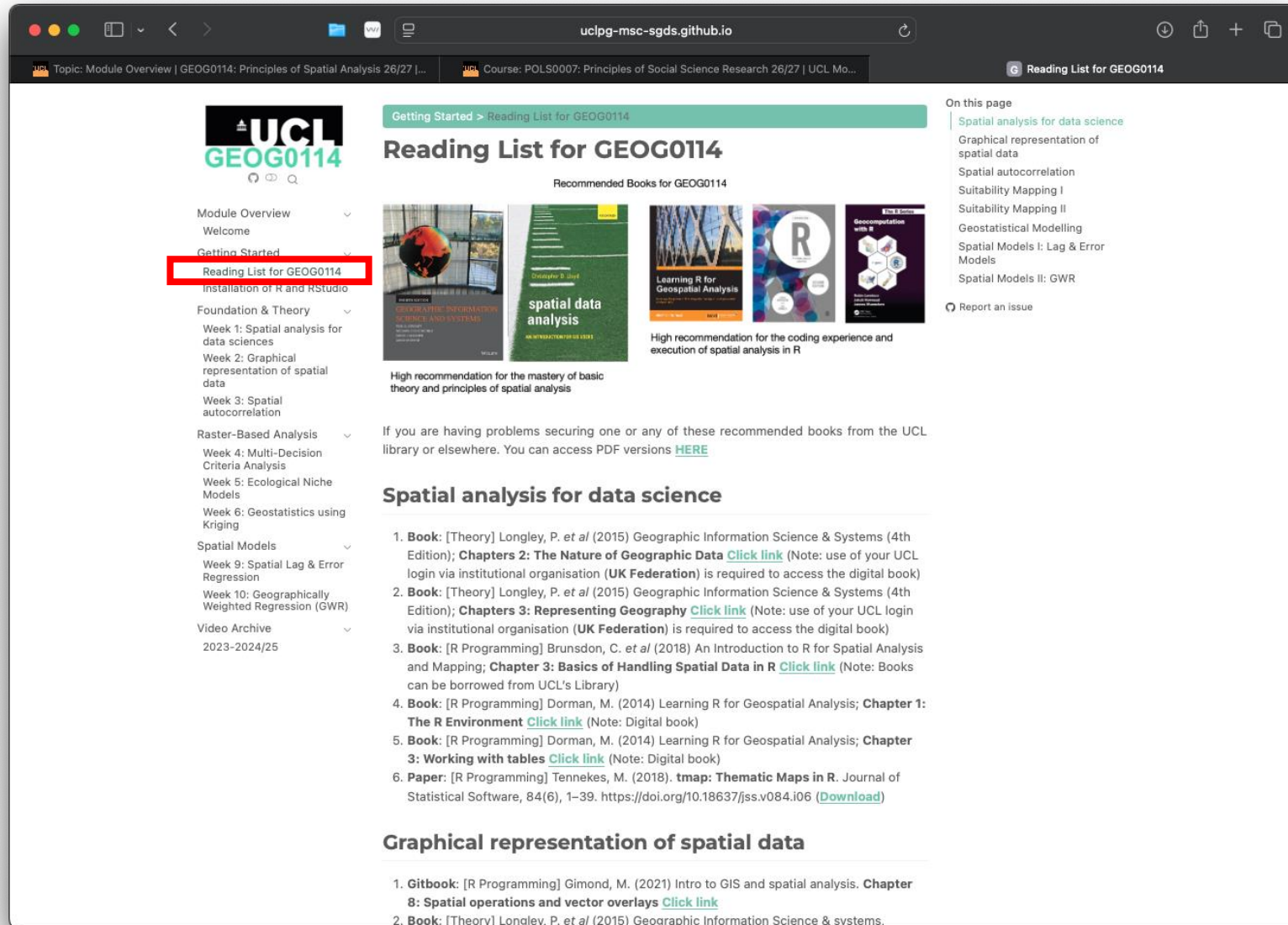
- Download the lecture and practical materials for that week

Recommended Books for GEOG0114



High recommendation for the coding experience and execution of spatial analysis in R

High recommendation for the mastery of basic theory and principles of spatial analysis



ucipg-msc-sgds.github.io

Topic: Module Overview | GEOG0114: Principles of Spatial Analysis 26/27 | UCL Mo... Course: POLS0007: Principles of Social Science Research 26/27 | UCL Mo... Reading List for GEOG0114

UCL GEOG0114

Module Overview
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Installation of R and RStudio
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Video Archive
2023-2024/25

Getting Started > Reading List for GEOG0114

Reading List for GEOG0114

Recommended Books for GEOG0114

High recommendation for the mastery of basic theory and principles of spatial analysis

If you are having problems securing one or any of these recommended books from the UCL library or elsewhere. You can access PDF versions [HERE](#)

Spatial analysis for data science

- Book:** [Theory] Longley, P. *et al* (2015) Geographic Information Science & Systems (4th Edition); **Chapters 2: The Nature of Geographic Data** [Click link](#) (Note: use of your UCL login via institutional organisation (**UK Federation**) is required to access the digital book)
- Book:** [Theory] Longley, P. *et al* (2015) Geographic Information Science & Systems (4th Edition); **Chapters 3: Representing Geography** [Click link](#) (Note: use of your UCL login via institutional organisation (**UK Federation**) is required to access the digital book)
- Book:** [R Programming] Brunson, C. *et al* (2018) An Introduction to R for Spatial Analysis and Mapping; **Chapter 3: Basics of Handling Spatial Data in R** [Click link](#) (Note: Books can be borrowed from UCL's Library)
- Book:** [R Programming] Dorman, M. (2014) Learning R for Geospatial Analysis; **Chapter 1: The R Environment** [Click link](#) (Note: Digital book)
- Book:** [R Programming] Dorman, M. (2014) Learning R for Geospatial Analysis; **Chapter 3: Working with tables** [Click link](#) (Note: Digital book)
- Paper:** [R Programming] Tennekens, M. (2018). **tmap: Thematic Maps in R**. Journal of Statistical Software, 84(6), 1–39. <https://doi.org/10.18637/jss.v084.i06> ([Download](#))

Graphical representation of spatial data

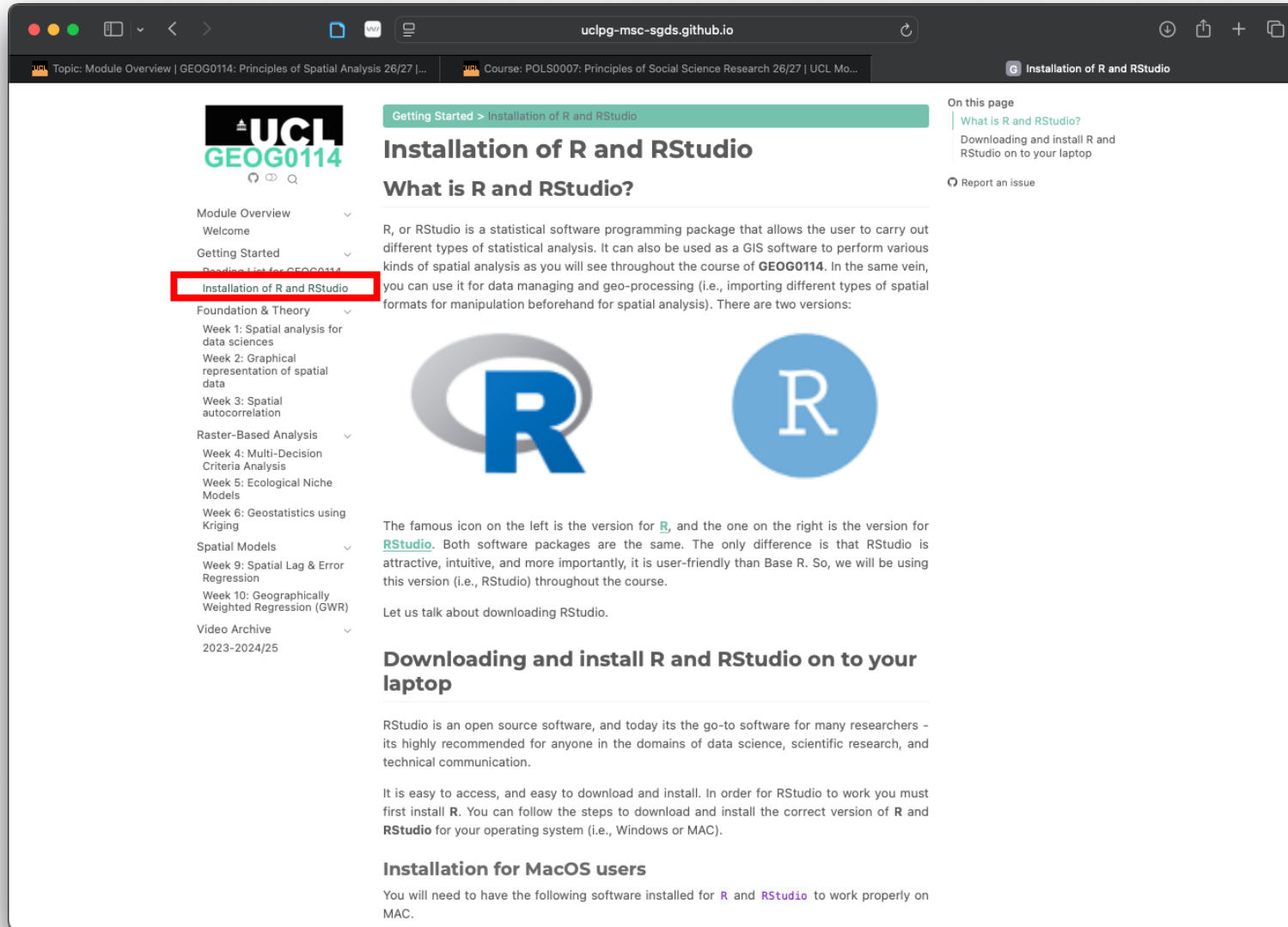
- Gitbook:** [R Programming] Gimond, M. (2021) Intro to GIS and spatial analysis. **Chapter 8: Spatial operations and vector overlays** [Click link](#)
- Book:** [Theory] Longley, P. *et al* (2015) Geographic Information Science & systems,

On this page

- Spatial analysis for data science
- Graphical representation of spatial data
- Spatial autocorrelation
- Suitability Mapping I
- Suitability Mapping II
- Geostatistical Modelling
- Spatial Models I: Lag & Error Models
- Spatial Models II: GWR

[Report an issue](#)

- You get the PDF versions the recommended textbooks, and all research papers under the 'Reading list for GEOG0114' section.



The screenshot shows a web browser window displaying the UCL GEOG0114 course page. The page title is "Installation of R and RStudio". The left sidebar contains a navigation menu with the following items: Module Overview, Welcome, Getting Started, **Installation of R and RStudio** (highlighted with a red box), Foundation & Theory, Raster-Based Analysis, Spatial Models, and Video Archive. The main content area has a green header bar with "Getting Started > Installation of R and RStudio". Below this, the section "What is R and RStudio?" explains that R or RStudio is a statistical software programming package. It features two logos: the R logo (a blue 'R' inside a grey circle) and the RStudio logo (a blue circle with a white 'R'). The text states that both packages are the same, but RStudio is more user-friendly. Below this, the section "Downloading and install R and RStudio on to your laptop" explains that RStudio is an open source software and provides instructions for downloading and installing it. The final section, "Installation for MacOS users", mentions that certain software must be installed for R and RStudio to work properly on MAC.

- All resources for downloading R and RStudio software (latest versions) are under the 'Installation of R and RStudio' section.

ucplg-msc-sgds.github.io

Topic: Module Overview | GEOG0114: Principles of Spatial Analysis 26/27 | ... Course: POLS0007: Principles of Social Science Research 26/27 | UCL Mo... Week 1: Spatial analysis for data sciences

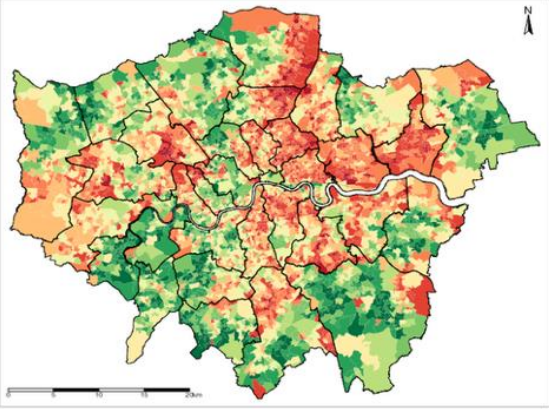
UCL GEOG0114

Module Overview

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 - Week 1: Spatial analysis for data sciences**
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 - Week 9: Spatial Lag & Error Regression
 - Week 10: Geographically Weighted Regression (GWR)
- Video Archive
 - 2023-2024/25

Note

Geography 101 – when creating a map, it is always best to add the North compass and scale bar. This is done with the `tm_compass()` and `tm_scalebar()` functions. The `tm_layout()` allows you to make further customisations to the internal and external plot region of map such as turning of the plot frame or modifying the text size in output. You can experiment with them by checking the help menu – just type: `?tm_layout()` in the console.



This concludes this computer lab session. Challenge yourself with the task placed in **section 1.6**

4 References

4.1 Recommended reading (see reading list)

1. **Book:** [Theory] Longley, P. et al (2015) Geographic Information Science & Systems (4th Edition); **Chapters 2: The Nature of Geographic Data** [Click link](#) (Note: use of your UCL login via institutional organisation (**UK Federation**) is required to access the digital book)
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4. **Book:** [R Programming] Dorman, M. (2014) Learning R for Geospatial Analysis; **Chapter 1:**

On this page

- 1 Introduction
- 2 Basics of managing data in RStudio
- 3 Basic visualisation of spatial data in RStudio
 - 3.1 Installing packages into RStudio
 - 3.2 Adding shapefiles in RStudio
 - 3.3 Mapping with 'tmap' functions in RStudio
 - 3.3.1 Visualising the outline(s) of study area
 - 3.3.2 Adding another layer on top of study area
 - 3.3.3 Full visualising of data in the maps
- 4 References
- 5 Data Sources
- 6 Exercise

[Report an issue](#)

- You can access computer practicals via the weekly chapters on the left-side of the screen
- You can navigate within that chapter using the page select on the right-side of the screen

What is Spatial Statistics (or Analysis)?

Definition:

The field of Spatial Statistics (or Analysis) is built on the assertion that nearby geographical observations (or objects) are somewhat associated in some way in space.

- The field is interdisciplinary as it brings together theories from **statistics, geography, computer science** and integrates **evidence-based research** methodologies (i.e., study designs)
- Usage for describing spatial distribution of **areal/point/gridded** outcomes, **interactions** between objects in space, but also how **an object has an impact on other nearby objects**.

Traditional statistics versus spatial statistics [1]

How does traditional statistics differ from spatial analysis?

The benchmark in traditional statistical theory is the “**common**” assumption of **independence** about observations which states that:

“the value of an observation does not influence or affect the value of other observations”

Crude examples:

- Assessing the Body Mass Index (BMI) of new enrolled students of GEOG0114
- Average test scores from Year 10 mathematics classes from each school

Traditional statistics versus spatial statistics [2]

The field of spatial statistics is based on assumption of **dependence (or non-independence)** of observations

Spatial statistical theory of assumption of **dependence (or non-independence)** states that:

“...closer observations or nearby units in space are in someway closely related (i.e., similar in characteristics or interact more) than those that are distant from each other...”

The assertion is the “**cornerstone**” of spatial statistics. It is called:

The First Law of Geography (Tobler, 1979)

First Law of Geography [1]



Waldo Rudolph Tobler (1930 – 2018)

Tobler's First Law of Geography on “**everything is related to everything else, but near things are more related to distant things**”

This first law is the foundation of the fundamental concepts of spatial **dependence**. It is integrated to most of the families of models in spatial statistics

Very important concepts to keep in mind when you think about **spatial dependence**:

- **Spatial Autocorrelation**
- **Distance Decay**
- **Spatial Spillover**

What is **Spatial autocorrelation** in the context of Tobler's theory?

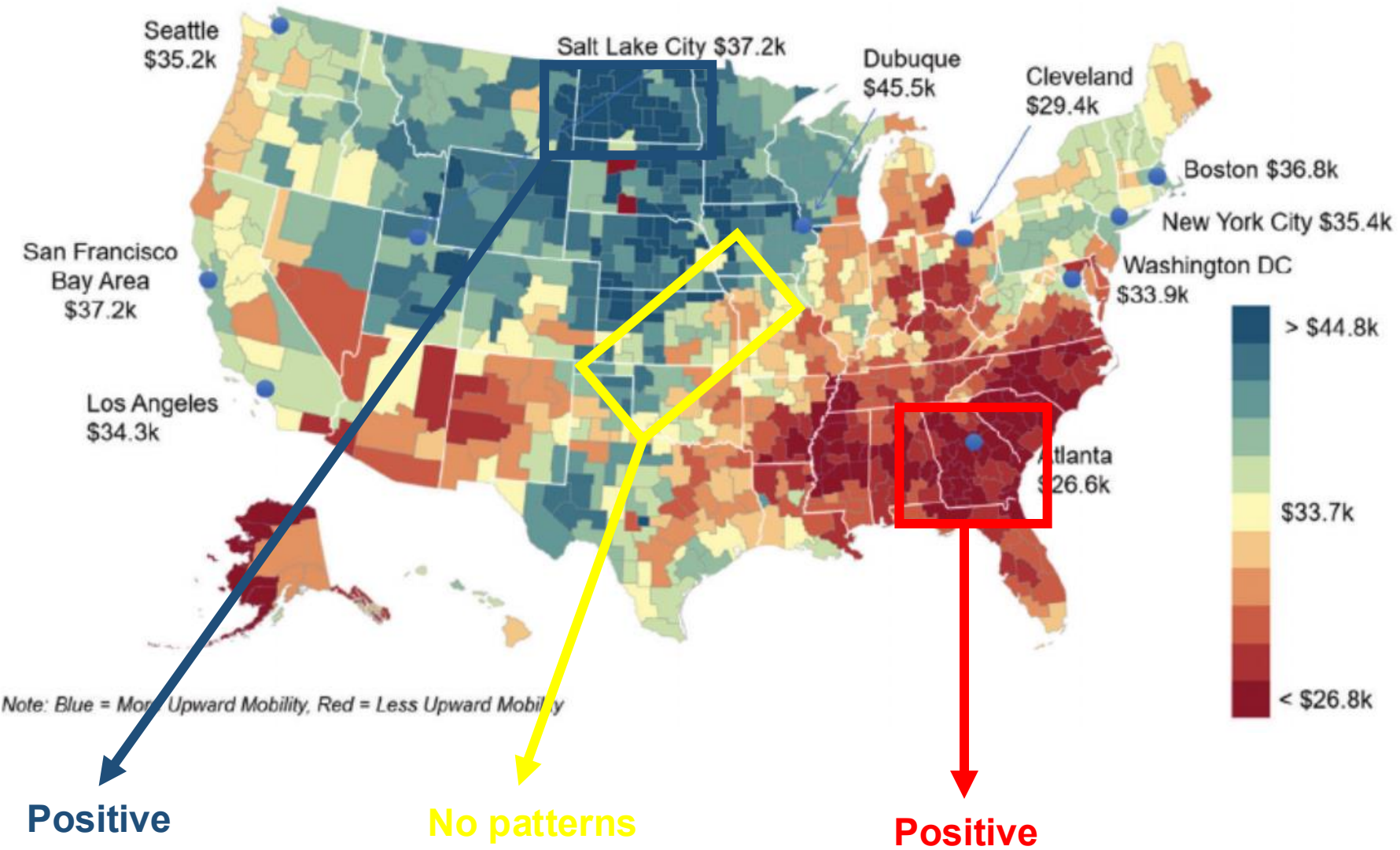
“This simply refers to whether (or not) similar values cluster together over geographic space”

Etymology (“**Auto**” means self; and “**correlation**” means the degree of relative correspondence)

Similar values that cluster together are said to have **positive spatial autocorrelation (& spatially dependent)**

Random patterns or values that cluster together are said to have **no spatial autocorrelation (and thus no dependence)**

The Geography of Upward Mobility in the United States: Average Household Income for Children with Parents Earning \$27,000 (25th percentile)



Spatial dependence and autocorrelation, particularly in areal data, will be covered in-depth in Week 3

What is **Distance decay** in the context of Tobler's theory?

“This refers to the distance between two events (or objects) and their degree of interaction in space as distance vary.”

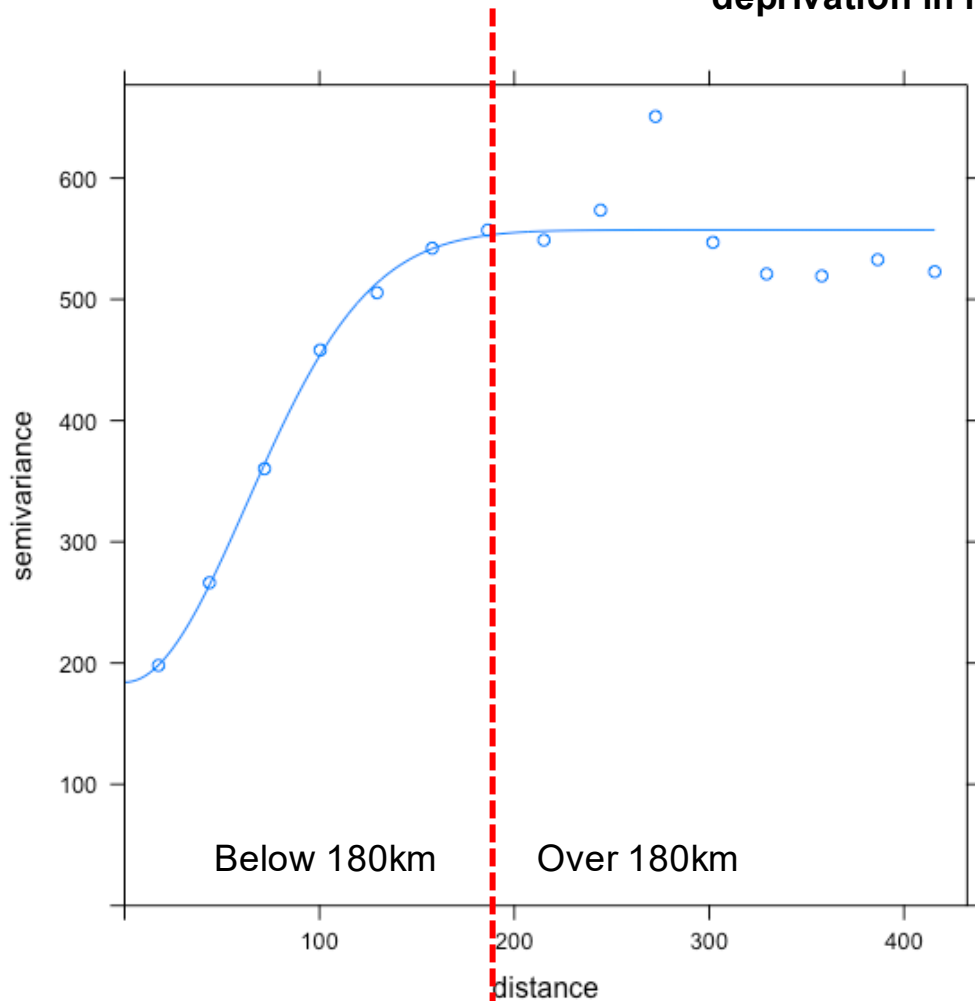
In short, what distance decay (in the context of Tobler's theory) says:

- It dictates how objects interact. The larger the distance between two events in a space – the less is their interaction & *vice versa*

Examples of where **Distance decay manifests in studies cities and urban space:**

- As distance from the focal point of a city center increases – high population density, taller buildings, and accessibility to multiple modes of transport decreases
- Housing marketing – house prices decreases, and in turn, residential mobility increases from expensive to affordable areas

Using Kriging to spatially predict areas with intense hookworm infection associated with socioeconomic deprivation in Northwestern Tanzania

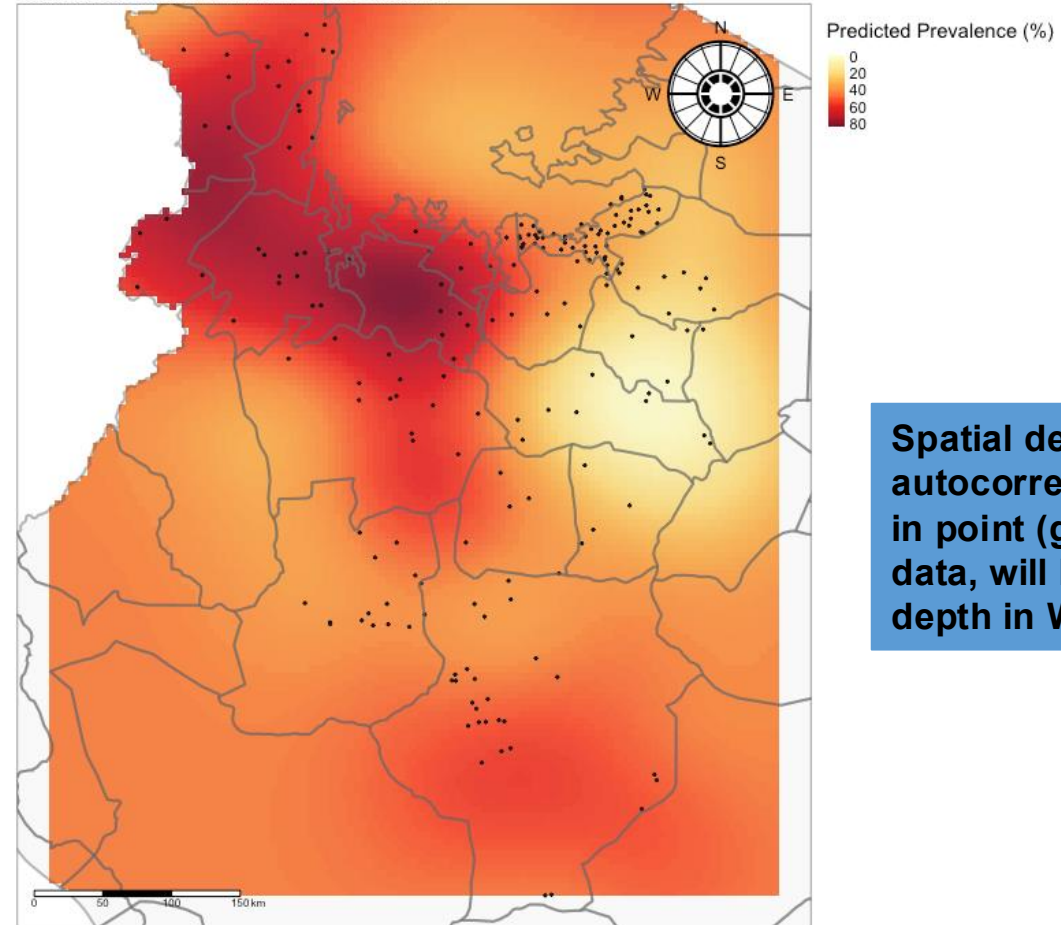


Villages with hookworm prevalence with separation distance below 180km have similar prevalence

Villages with hookworm prevalence with separation distance 180km and over don't have related prevalence

We used the concept of distance decay, or in this 'separation distance' to account for spatial dependence between our survey points

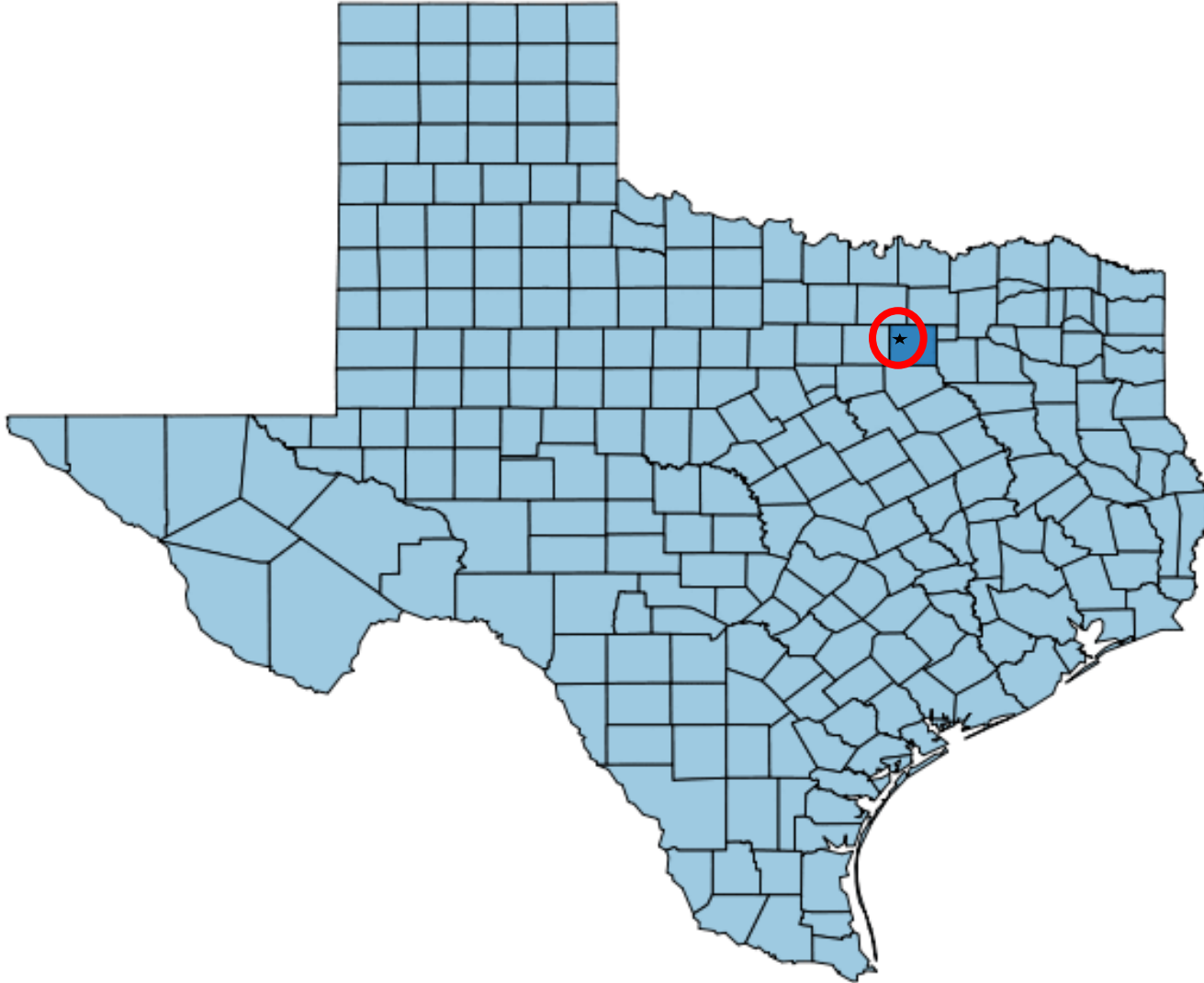
Hookworm Infection in North West Tanzania



Spatial dependence and autocorrelation, particularly in point (geo-statistical) data, will be covered in-depth in Week 6

We have survey points (of villages) reporting prevalence of hookworm in Northwestern Tanzania. We assumed there some spatial dependence in the prevalence of hookworm in these locations to predict prevalence where there are no information (or points). We use this information on form the left panel to build our model for making geostatistical predictions.

Spatial Spillover: “...where you are in geographic space matters” an event in one location can somehow have an impact on other events in neighboring areas

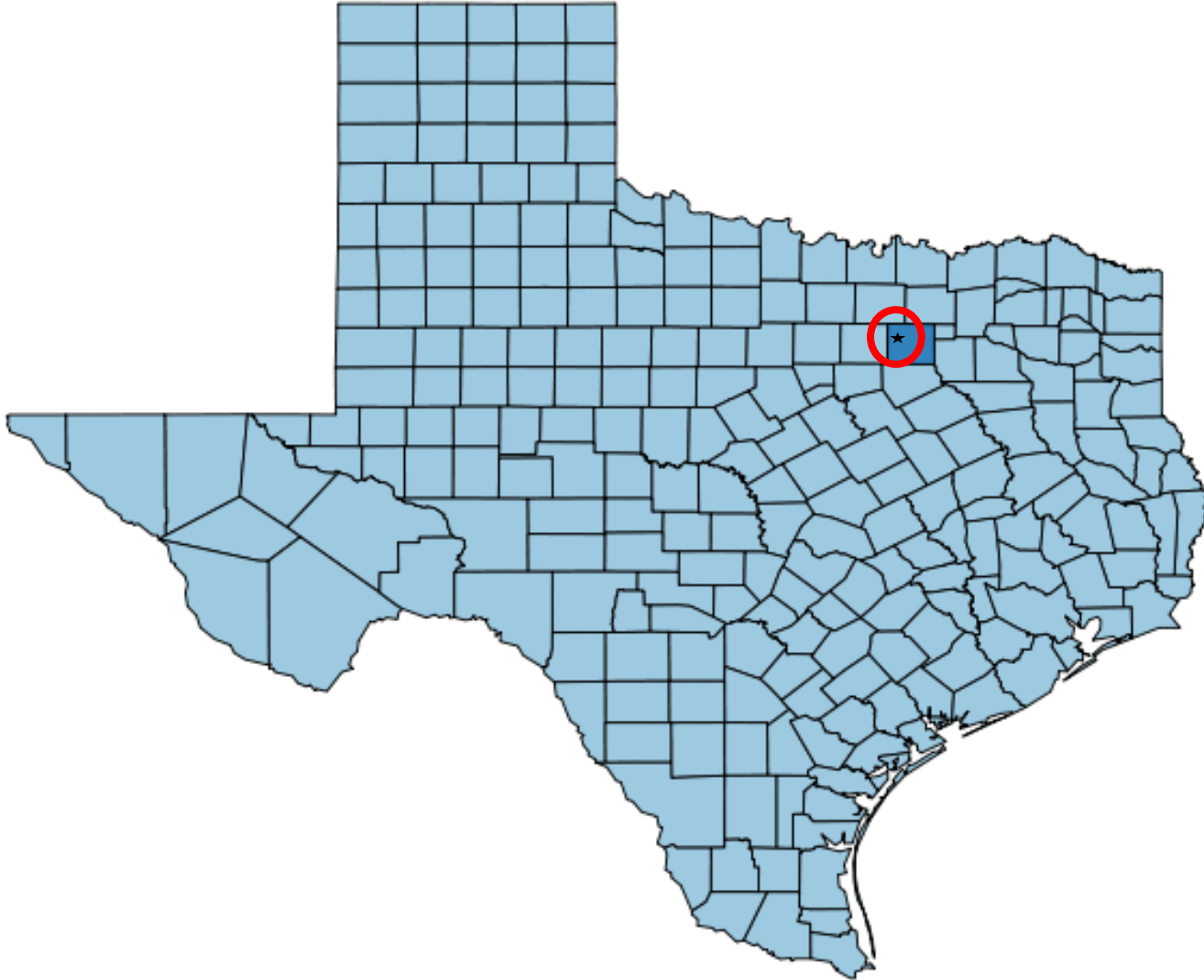


In Spatial Statistics, especially in Spatial Regression, Geostatistics – we often try to account for **spatial spillover** effects in our models

There are four types of spillovers:

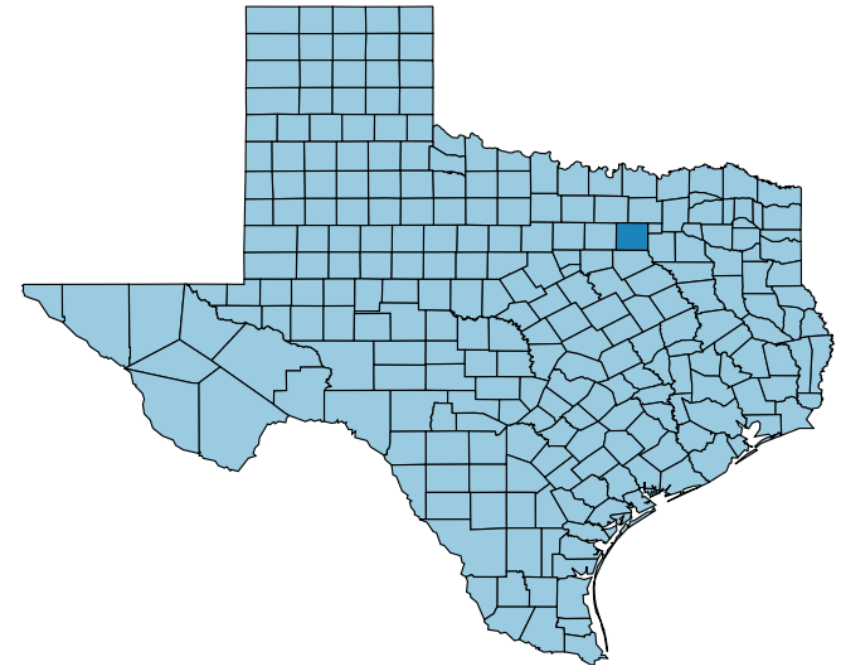
- No spillover
- Local spillover
- Global spillover
- Rippling spillovers

Spatial Spillover: “...where you are in geographic space matters” an event in one location can somehow have an impact on other events in neighboring areas



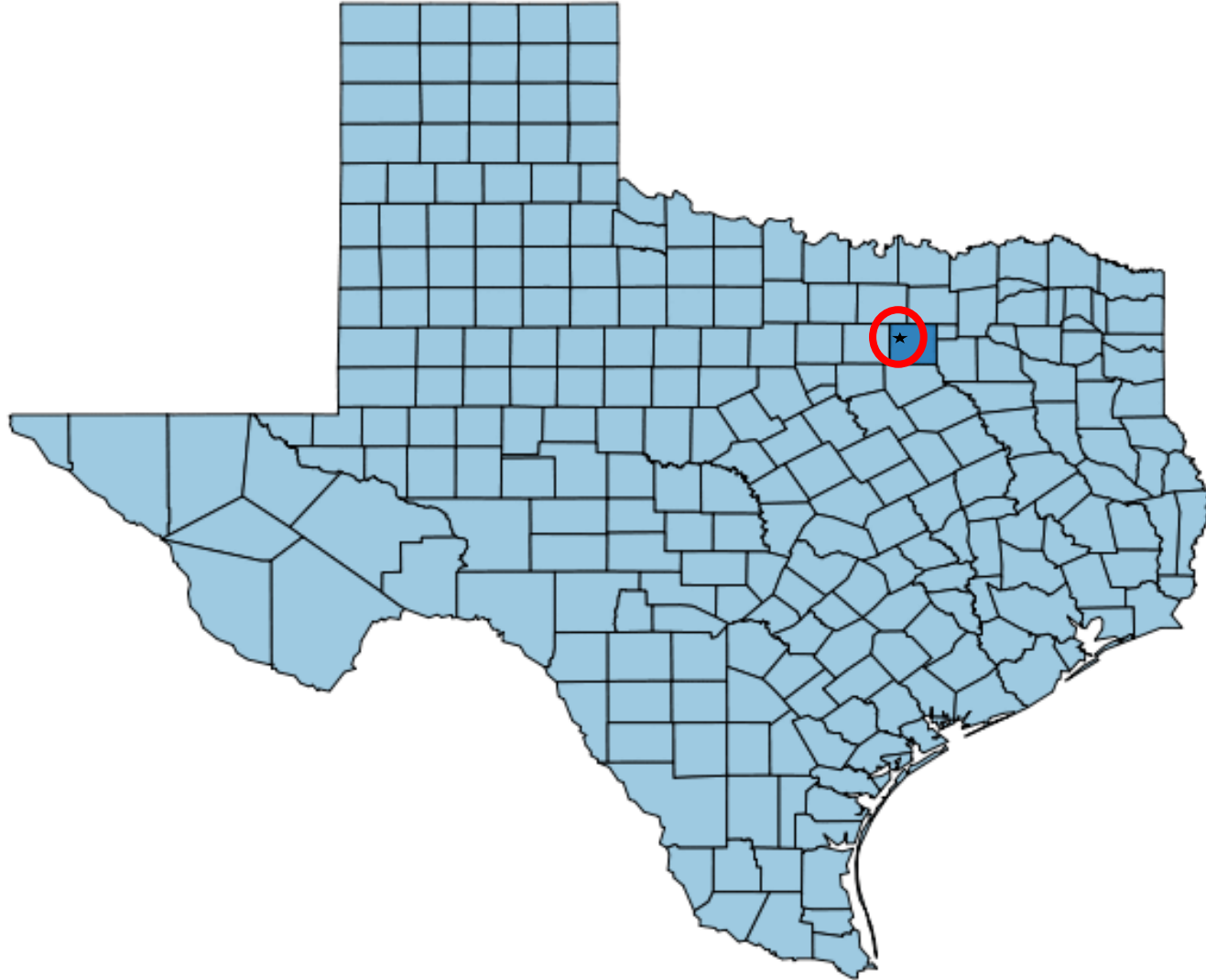
In Spatial Statistics, especially in Spatial Regression, Geostatistics – we often try to account for **spatial spillover** effects in our models

No spillover (independence)



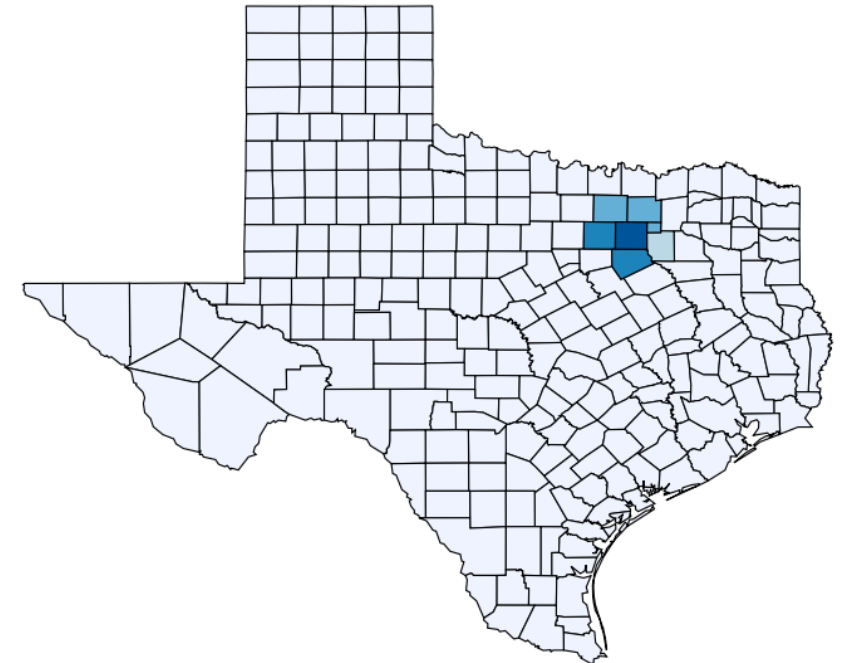
A change in unemployment in that dark blue area will only have an impact on crime rates within its own area, but does not influence crime rates in neighboring areas

Spatial Spillover: “...where you are in geographic space matters” an event in one location can somehow have an impact on other events in neighboring areas



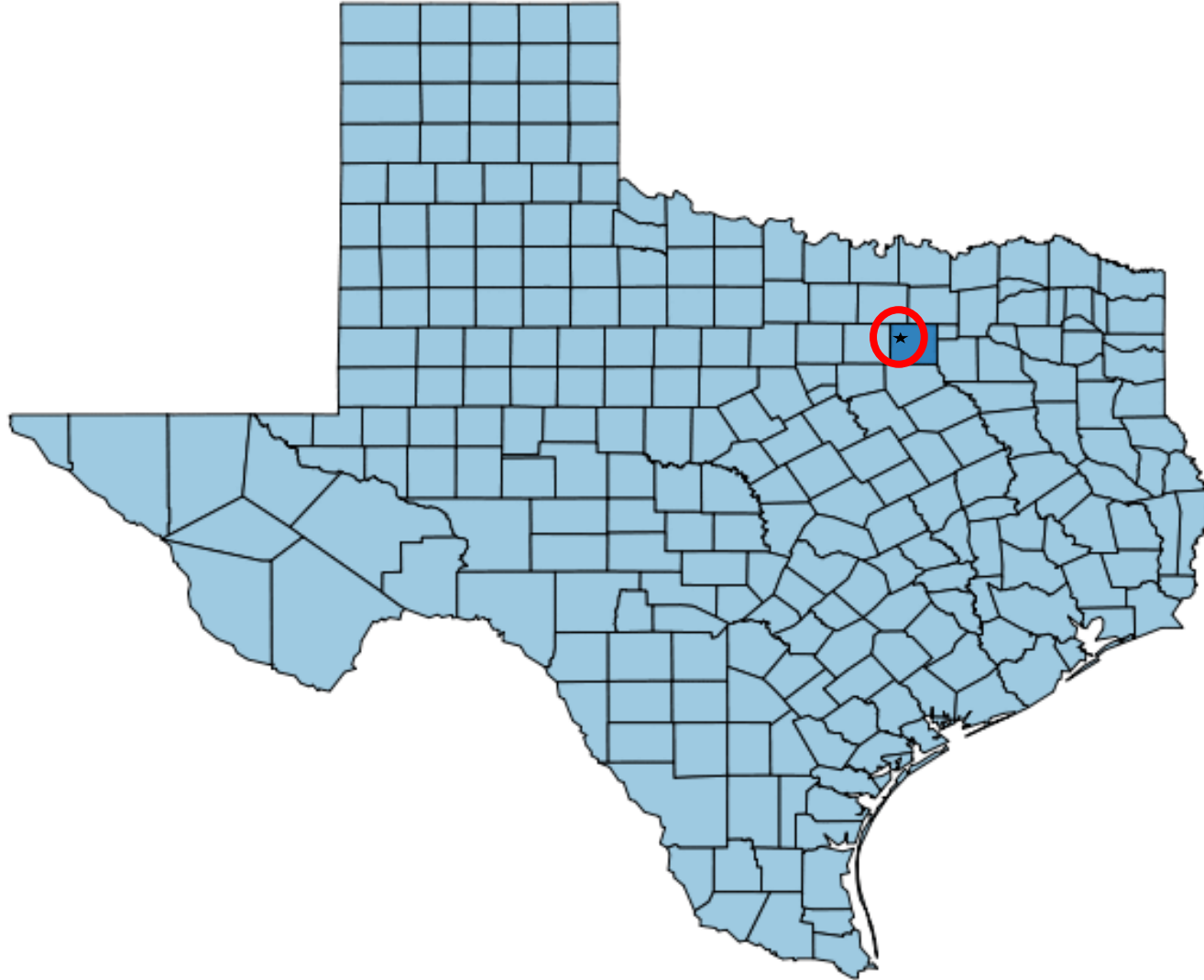
In Spatial Statistics, especially in Spatial Regression, Geostatistics – we often try to account for **spatial spillover** effects in our models

Local spillover (dependence)



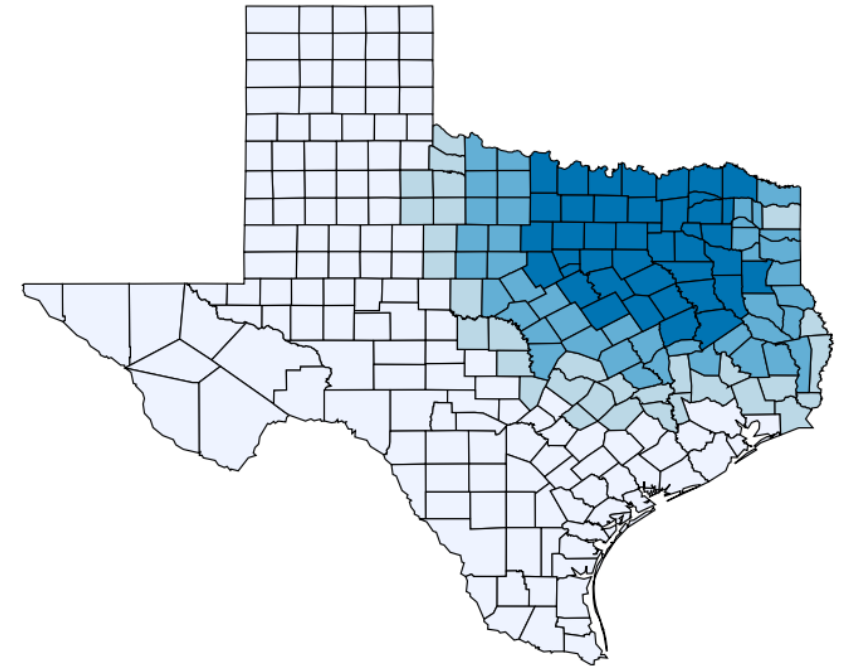
A change in unemployment in that dark blue area (in the center) will not only have an impact on crime rates within its own area, but it will also have a direct influence on crime rates in neighboring areas only.

Spatial Spillover: “...where you are in geographic space matters” an event in one location can somehow have an impact on other events in neighboring areas



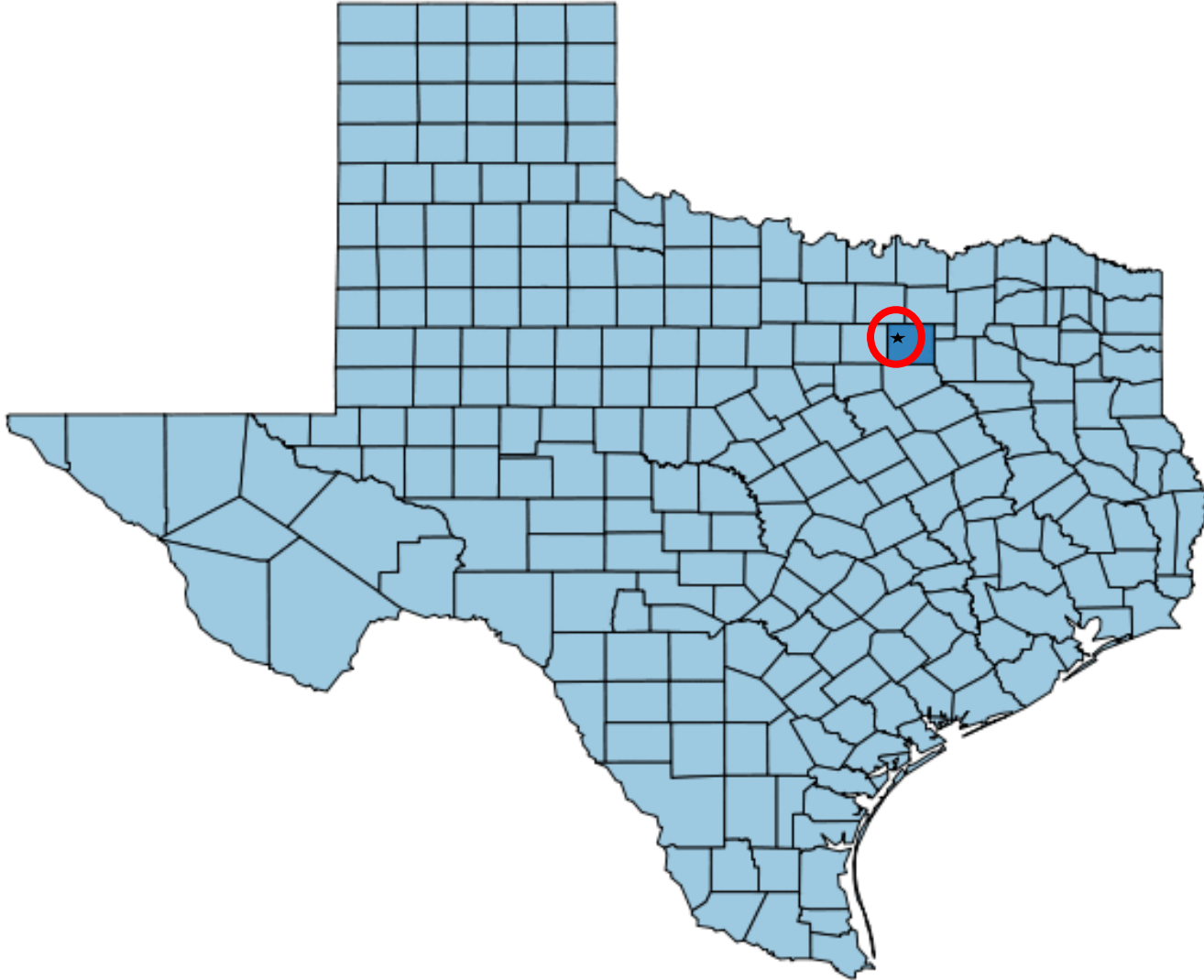
In Spatial Statistics, especially in Spatial Regression, Geostatistics – we often try to account for **spatial spillover** effects in our models

Global spillover (dependence)



A change in unemployment in that dark blue area (in the center) will not only have an impact on crime rates within its own area, but it will also have a wider influence on crime rates beyond its direct neighbors.

Spatial Spillover: “...where you are in geographic space matters” an event in one location can somehow have an impact on other events in neighboring areas



In Spatial Statistics, especially in Spatial Regression, Geostatistics – we often try to account for **spatial spillover** effects in our models

Rippling spillover (dependence)

Where there is a focal point for an event and its influence may have a rippling (or trickle down) effect across space, triggering other events, which then diminishes with time and distance.

e.g.,

- Natural disasters - an earthquake and building destruction.
- Disease spread and outbreaks
- World financial markets and crashes

Spatial dependence in spillovers, particularly in areal data from a modelling approach, will be covered in-depth in Week 9

What is Spatial Data & its Features?

Point Pattern Data (PPD)

Key Characteristics

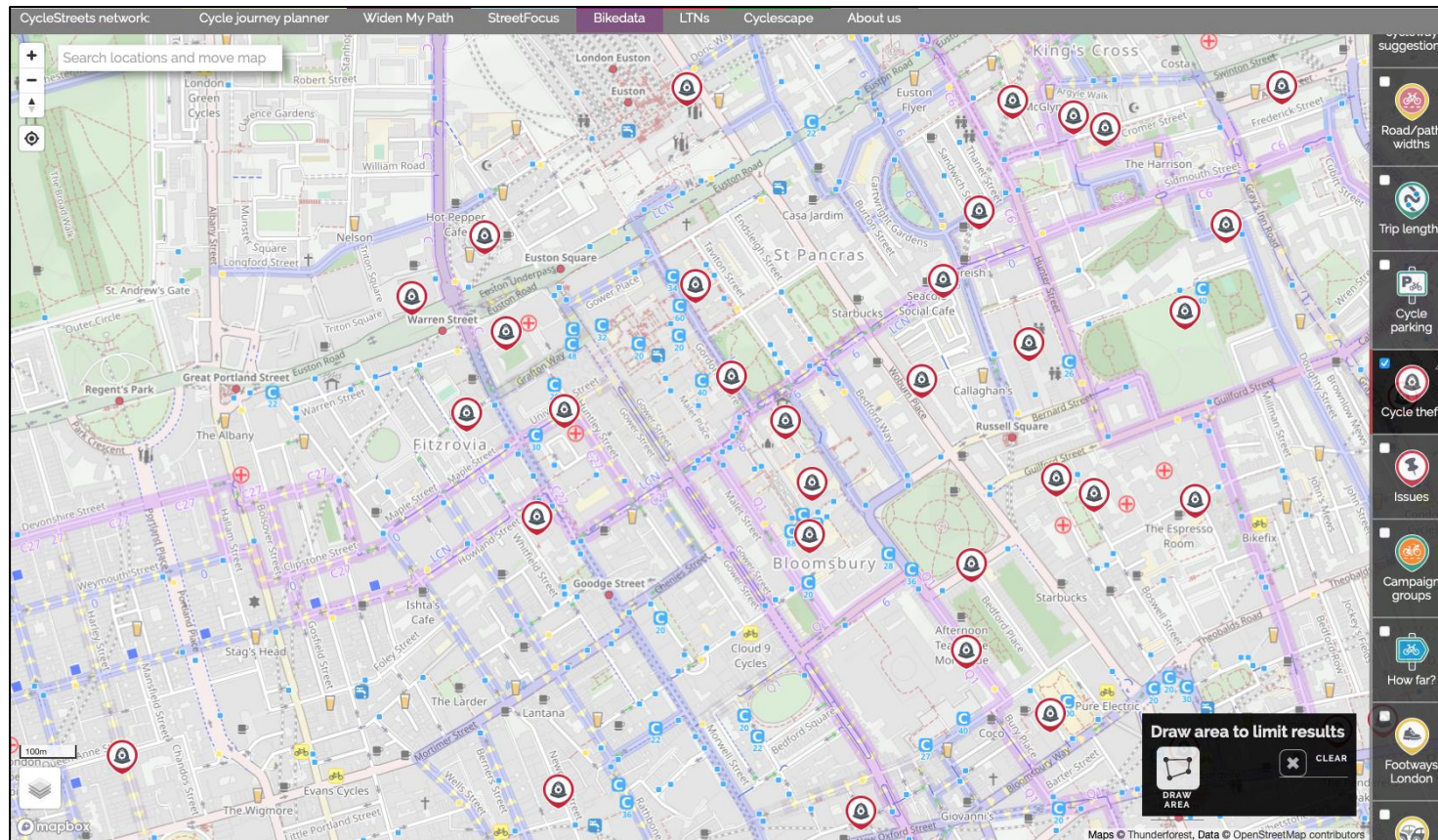
The main interest is the occurrences of an event at a points (or points). These events occur at “random” at any given geographic space and time.

Examples of PPDs events (or outcomes):

- Point locations of burglaries
- Riots
- Locations of car collisions etc.,
- Locations of where an adult tree needs to be replanted

Some PPDs events may carry additional information that may describe the occurrence of an observed event (or outcome):

- Burglary: Type of premise that was burgled, time of day the burglary occurred etc.,
- Car collision: type of road, weather condition etc.



Represents point locations of bicycle thefts in Central London area

Source: BikeData.CycleStreets network <https://bikedata.cyclestreets.net/>

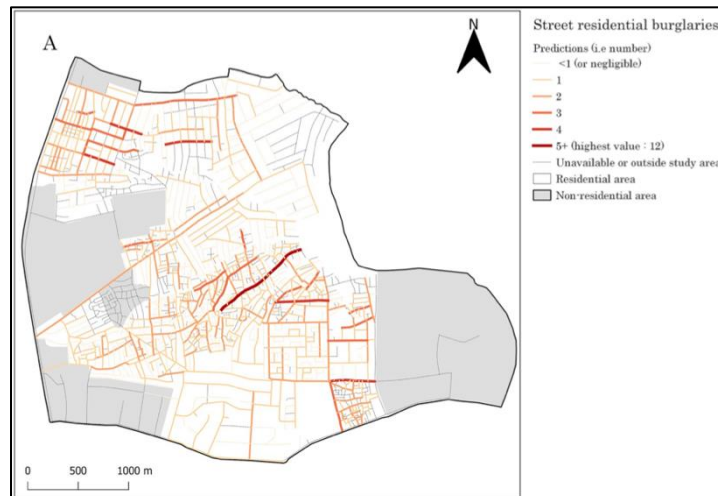
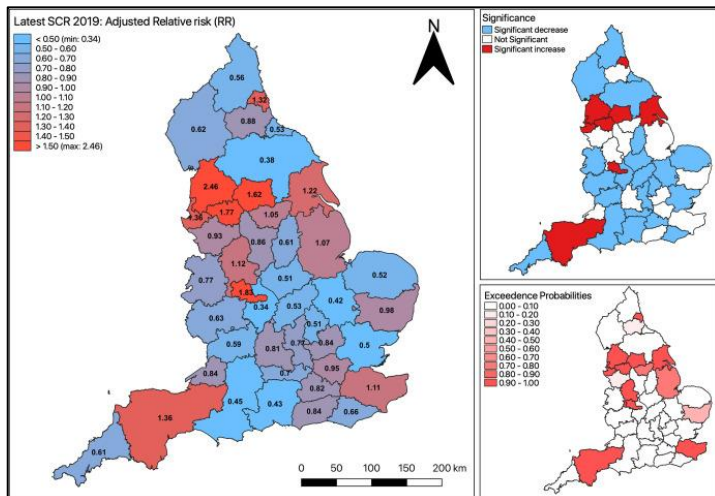
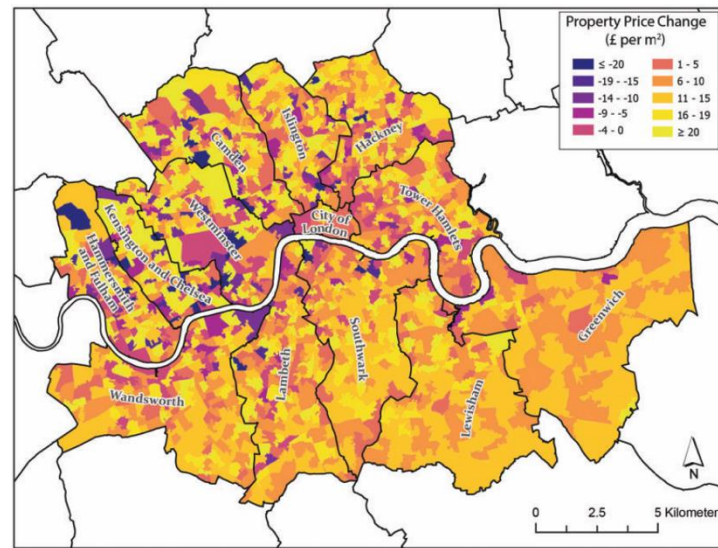
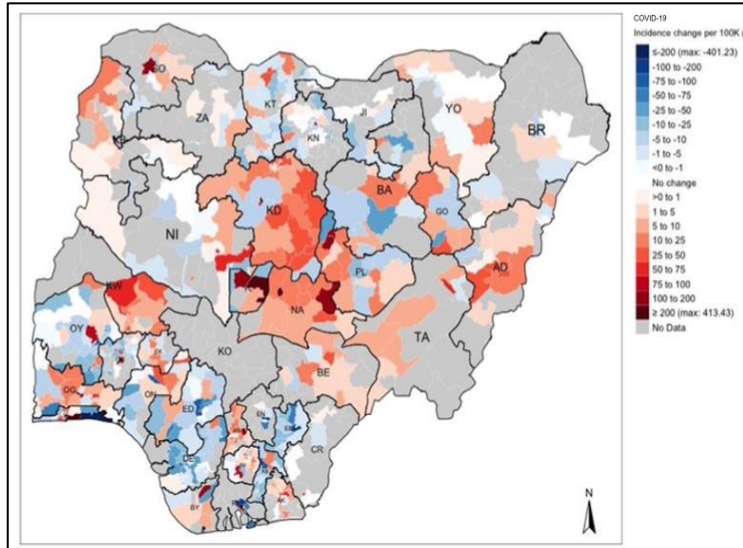
Aggregated Data

Key Characteristics

The main interest the quantity of interest defined for line segments, areas, or regions.

Events (or outcomes) that are aggregated measures to areas:

- Prevalence of a disease in areas
- Population density in a county
- Regional unemployment rates
- Risk of an outcome



Sources:

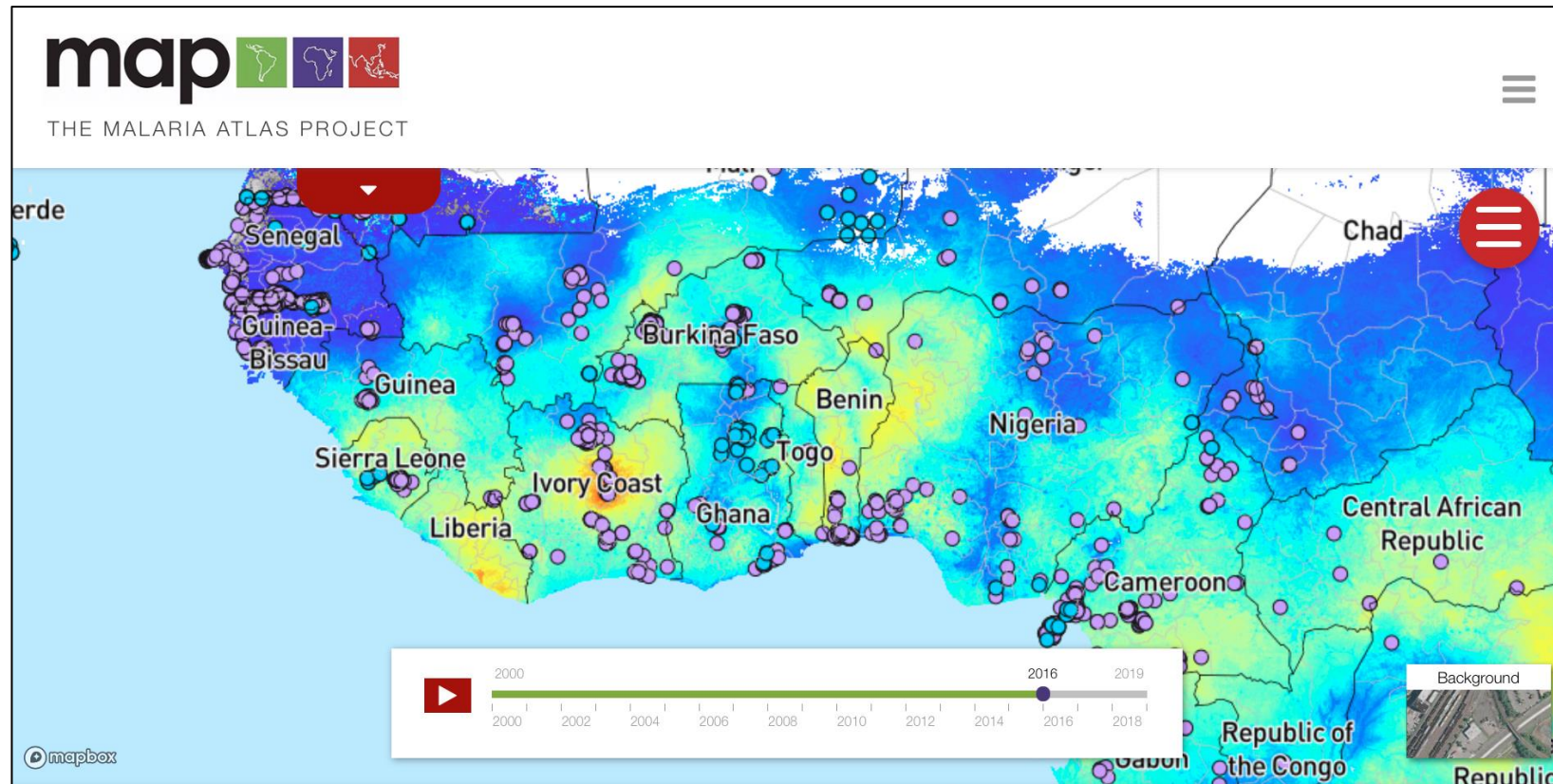
Musah, A., et al. (2020): <https://doi.org/10.1016/j.apgeog.2019.102126>

Todd, J., et al. (2021): <https://doi.org/10.1177/23998083211001836>

Li, L., et al. (2022): <https://doi.org/10.1016/j.apgeog.2022.102718>

Elimian, K., et al. (2022): <http://dx.doi.org/10.1136/bmjopen-2022-063703>

Geostatistical Data



Key Characteristics

- The quantity of interest has a value at any location across a given area.
- These are values over a grid/raster

Events are usual statistical prediction determined from “**sampled**” points with continuous data values:

- Land surface elevation
- Diffusion of ambient air pollutants
- Environmental suitability for breeding habitats of mosquitoes
- Community surveys pertained to disease burden

Sampled points are surveys on prevalence of malaria, which were used to make survey predication of prevalence at unsampled areas in Sub-Saharan Africa

Source: The Malaria Atlas Project <https://malariaatlas.org>

Non-spatial context & data structure:

	ATTRIBUTE			
	Variable 1	Variable 2	...	Variable n
Entity 1	$attribute_{11}$	$attribute_{12}$	...	$attribute_{1n}$
Entity 2	$attribute_{21}$	$attribute_{22}$	...	$attribute_{2n}$
⋮	⋮	⋮	⋮	⋮
Entity m	$attribute_{m1}$	$attribute_{m2}$	...	$attribute_{mn}$

To apply some spatial analysis to data – you must have some variable that defines the entity’s geographic location. This can be **GPS coordinates, spatially referenced geometries** of an area

NOTES: It is not enough to have the just the name of the area(s). It must be some geometric entry!

- Attributes that defines an entity’s location are typically excluded from the analysis
- The conventional statistical methods, that assumes independence, are used for analyzing such dataset
- Results churned from this dataset are completely independent from “**spatial arrangement**” of the entities.

Spatial context & data structure:

	Geographical Coordinate		ATTRIBUTE			
	X	Y	Variable 1	Variable 2	...	Variable n
Entity 1	X_1	Y_1	$attribute_{11}$	$attribute_{12}$	...	$attribute_{1n}$
Entity 2	X_2	Y_2	$attribute_{21}$	$attribute_{22}$	...	$attribute_{2n}$
⋮	⋮	⋮	⋮	⋮	⋮	⋮
Entity m	X_m	Y_m	$attribute_{m1}$	$attribute_{m2}$	...	$attribute_{mn}$

In this example, what defines the entity's geographic location are **X, Y GPS coordinates**.

Definition of an entity's location are not limited to coordinates, you can have spatially reference areas with their associated boundaries with geometries.

This instance illustrates an example of geostatistical data.

- Attributes that defines an entity's location are typically explicitly incorporated into analysis
- Spatial statistical methods, that assumes dependence, are used for analyzing such geographically referenced dataset
- Results churned from this dataset are completely dependent from “**spatial arrangement**” of the entities.

What is Geographical Information Systems (GIS)



Roger F. Tomlinson (1933 – 2014)

He is an English geographer and computer scientist, who is revered as the “**Father of GIS**”

- His PhD research titled: “*The application of electronic computing methods and techniques to the storage, compilation, and assessment of mapped data*” (1962, @UCL)
- Conceived the idea of analyzing multiple layers of spatial data within a single environment
- **Geographical Information Systems (GIS)**, which is now a software, grown into a billion-dollar industry







R (Standard)

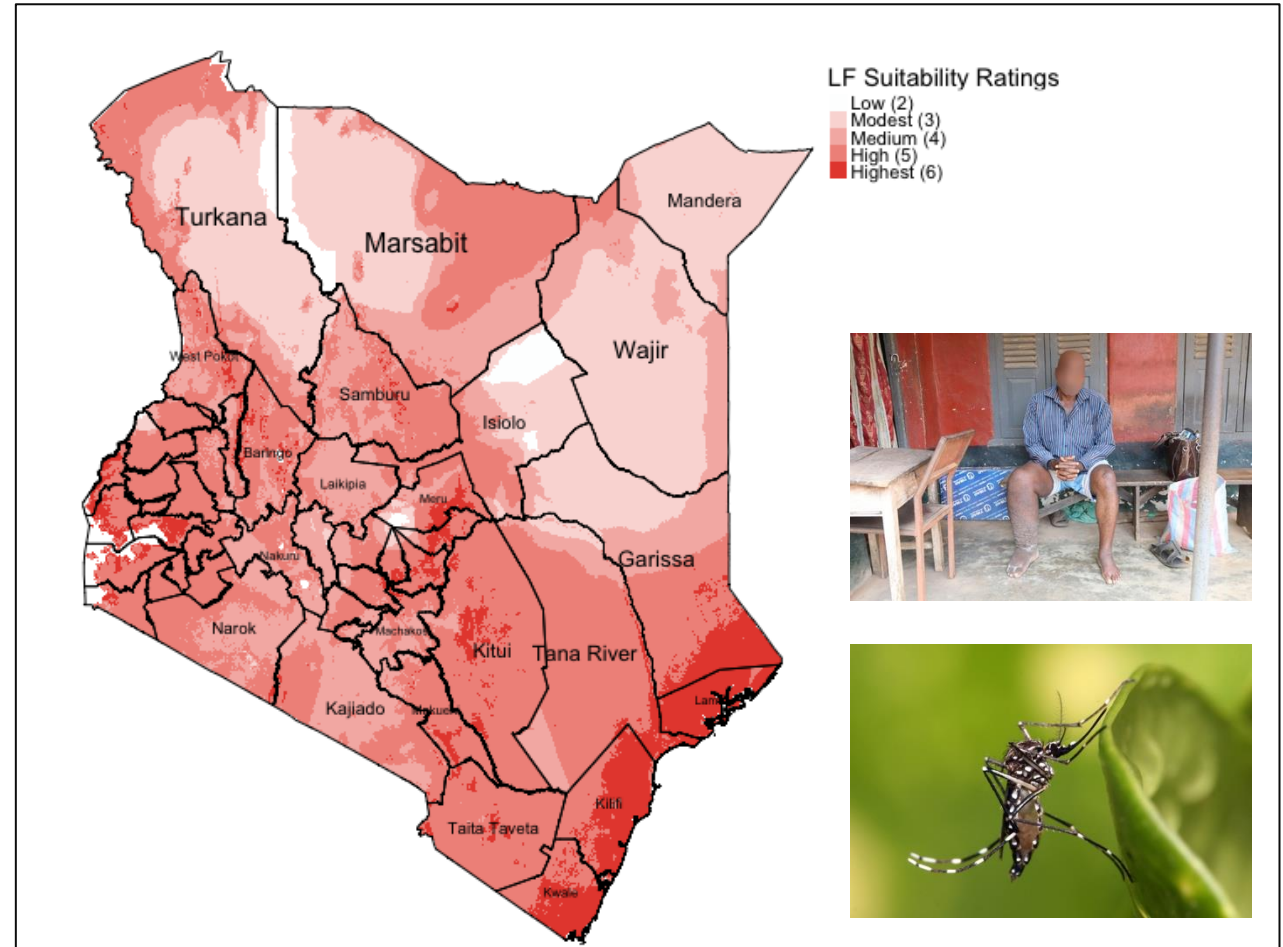


RStudio (Best)

There are two version of the software: 1.) R, and 2.) **RStudio**; The second is much preferred as its straightforward and intuitive.

Why are we teaching RStudio?

1. Flexible and provides access to powerful packages for analysis
2. Impressive graphs, visualizations and maps
3. Excellent statistical capabilities too



Example: Map generated in R to illustrate areas that are environmentally suitable for the spread of neglected tropical disease called 'Lymphatic Filariasis (LF)' in Kenya.

Sources:

1. Global Atlas for Helminths Infection (<http://www.thiswormyworld.org>)
2. ESPEN (<https://espen.afro.who.int>)

... and why learn how to code in RStudio?

1. Efficiency

- Automated tasks and data managing
- Can recycle & reuse code scripts for new projects

2. Fosters good scientific practice

- Transparency and replication (AKA reproducible research)
- Creates log so anyone can follow in your footsteps (i.e., github, gitlab etc.,)

You can literally pull-off some really creative stuff like generating websites, accessing tools via APIs etc.

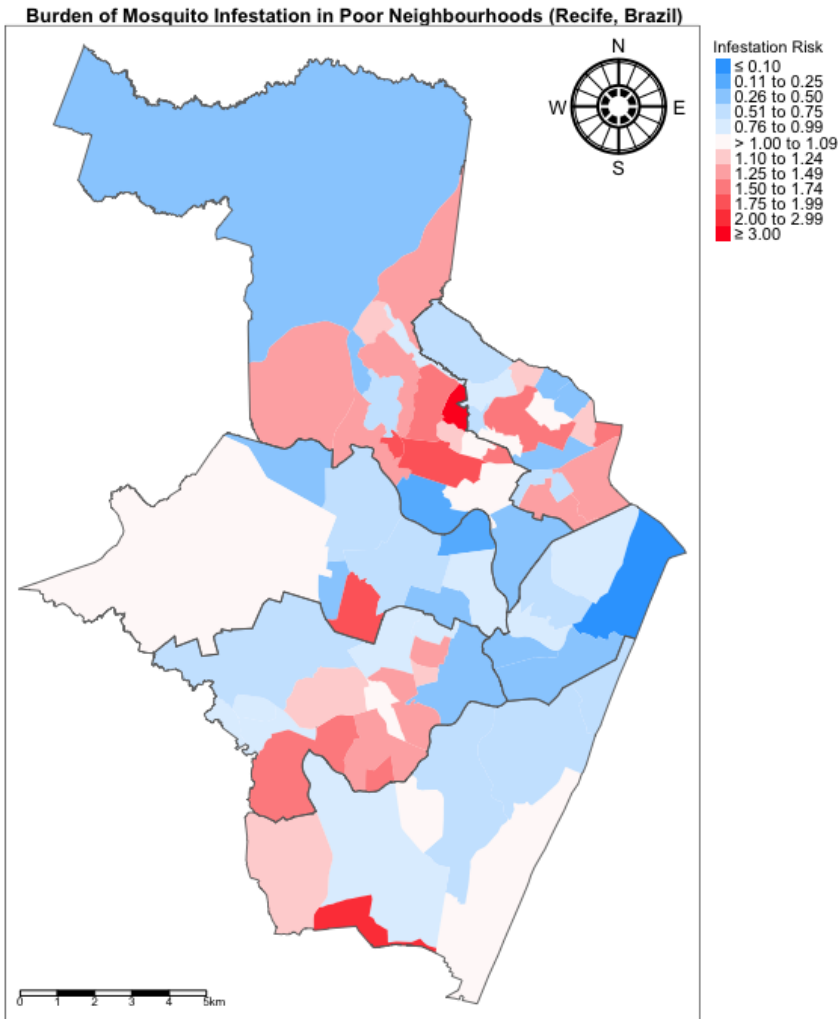
The screenshot shows the RStudio interface with a GitHub repository named 'UCLPG-MSC-SGDS / GEOG0114-PSA-WK8'. The left sidebar displays a file explorer with various files and folders, including '_site', 'docs', 'index_cache/html', '.DS_Store', '.gitignore', 'Data analysis script f...', 'Datasets.zip', 'GEOG0114-PSA-WK...', 'Landslide.zip', '_site.yml', 'ahp.Rmd', 'classification.Rmd', 'index.Rmd', 'landslide.Rmd', and 'recommendations.Rmd'. The main window shows the 'Code' tab with R code for creating a raster layer and visualizing it. The code includes comments and function calls like 'tm_shape', 'tm_raster', 'tm_polygons', and 'tm_scale_bar'. Below the code, there is a map of Kenya showing suitability levels, with regions labeled like Turkana, Marsabit, Mandera, Wajir, Garissa, Kilifi, and Tana River. An 'IMPORTANT NOTES' box at the bottom states: 'The suitability ranges are estimated to be from 3.27 to 7.49 (weighted on a scale with an upper limit of 10). The highest suitability for LF are regions with values closest to 7.49 and vice versa.'

Example: Working in RStudio and synchronising it with GitHub to not only use as a cloud back-up, but to generate a website through RStudio and GitHub for teaching MSc Students.

Sources:

1. GitHub (<https://github.com>)

Example of a basic code structure in RStudio



```
# comment: activate packages for performing GIS in R
library("sf")
library("tmap")

# comment: add neighbourhood shapefile w/mosquito infestation data using read_sf()
recife.neighbourhoods <- read_sf("Recife_neighb_epsg3857_fixed.shp")
recife.healthzone <- read_sf("Recife_regions_epsg3857_fixed.shp")

# comment: assigning labels for the risk estimate legends
RiskCategorylist <- c("\u2264 0.10", "0.11 to 0.25", "0.26 to 0.50", "0.51 to 0.75",
"0.76 to 0.99", ">1.00 to 1.09", "1.10 to 1.24", "1.25 to 1.49", "1.50 to 1.74", "1.75
to 1.99", "2.00 to 2.99", "\u2265 3.00")

# comment: generating the divergent color scheme from Blues to Red spectrum
RRPalette <- c("#33a6fe", "#65baf6", "#98cffe", "#cbe6fe", "#dfeffe", "#fef9f9",
"#fed5d5", "#feb1b1", "#fe8e8e", "#fe6a6a", "#fe4646", "#fe2424", "#fe0000")

# comment: map of risk of infestation
tm_shape(recife.neighbourhoods) +
  tm_fill("RelativeRiskCat",
    style = "cat",
    title = "Infestation Risk",
    palette = RRPalette,
    labels = RiskCategorylist) +
tm_shape(recife.healthzone) +
  tm_polygons(alpha = 0, border.alpha = 0.90) +
  tm_layout(frame = TRUE,
    main.title = "Mosquito Infestation in Neighbourhoods (Brazil)",
    main.title.size = 0.8,
    main.title.position = 0.02,
    main.title.fontface = 2,
    legend.outside = TRUE,
    legend.outside.position = "right",
    legend.title.size = 0.8,
    legend.text.size = 0.7) +
  tm_scale_bar(position = c("left", "bottom")) +
  tm_compass(type = "radar", show.labels = 2, position = c("right", "top"))
```

Summary

The take home message is:

- Spatial statistics is grounded on spatial dependence, drawn from **Tobler's 1st Law of Geography**
- Fundamental concepts for explaining spatial dependence – are three key words: **Spatial autocorrelations & Distance decay** effect, and spatial spillovers.
- Spatial data and data structure – three types: areal/regional data, point patterns and geostatistical
- Two important people to commit to memory: Dr Waldo Tobler (1st Law of Geography) and Dr Roger Tomlinson (Father of GIS)
- Best GIS packages are usual the ones that are FREE (i.e., R/RStudio for its packages that enable GIS functionality, and QGIS)

Any questions?

